Hands-on: Advanced functionals

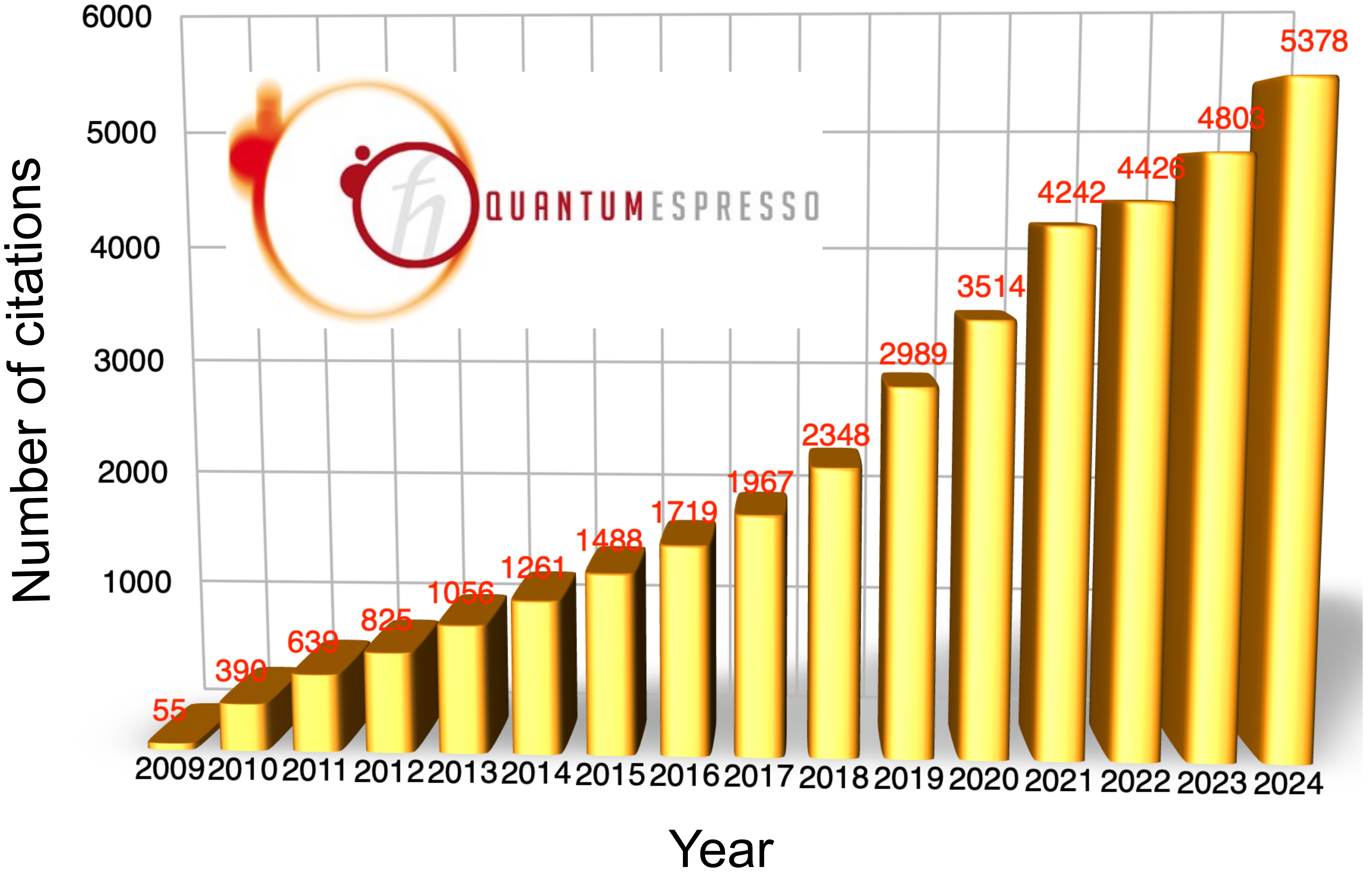
Presenters: Valentina Sanella (PSI) and Iurii Timrov (PSI)

open-source

QUANTUM ESPRESSO

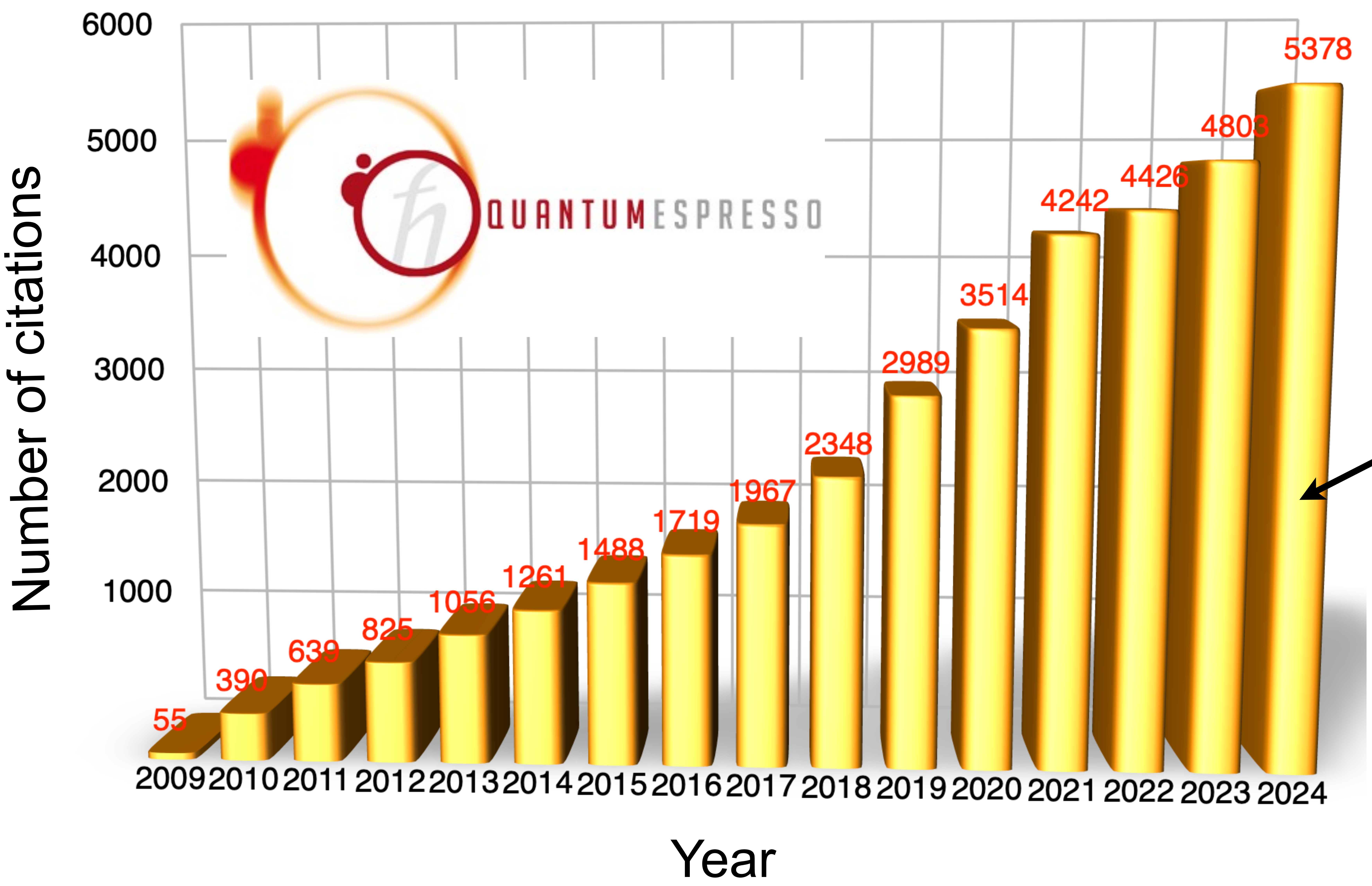
ELECTRONIC-STRUCTURE CODE
FOR MATERIALS SCIENCE APPLICATIONS

Quantum ESPRESSO



P. Giannozzi, S. Baroni *et al.*, J. Phys.: Condens. Matter 21, 395502 (2009); *ibid.* 29, 465901 (2017).

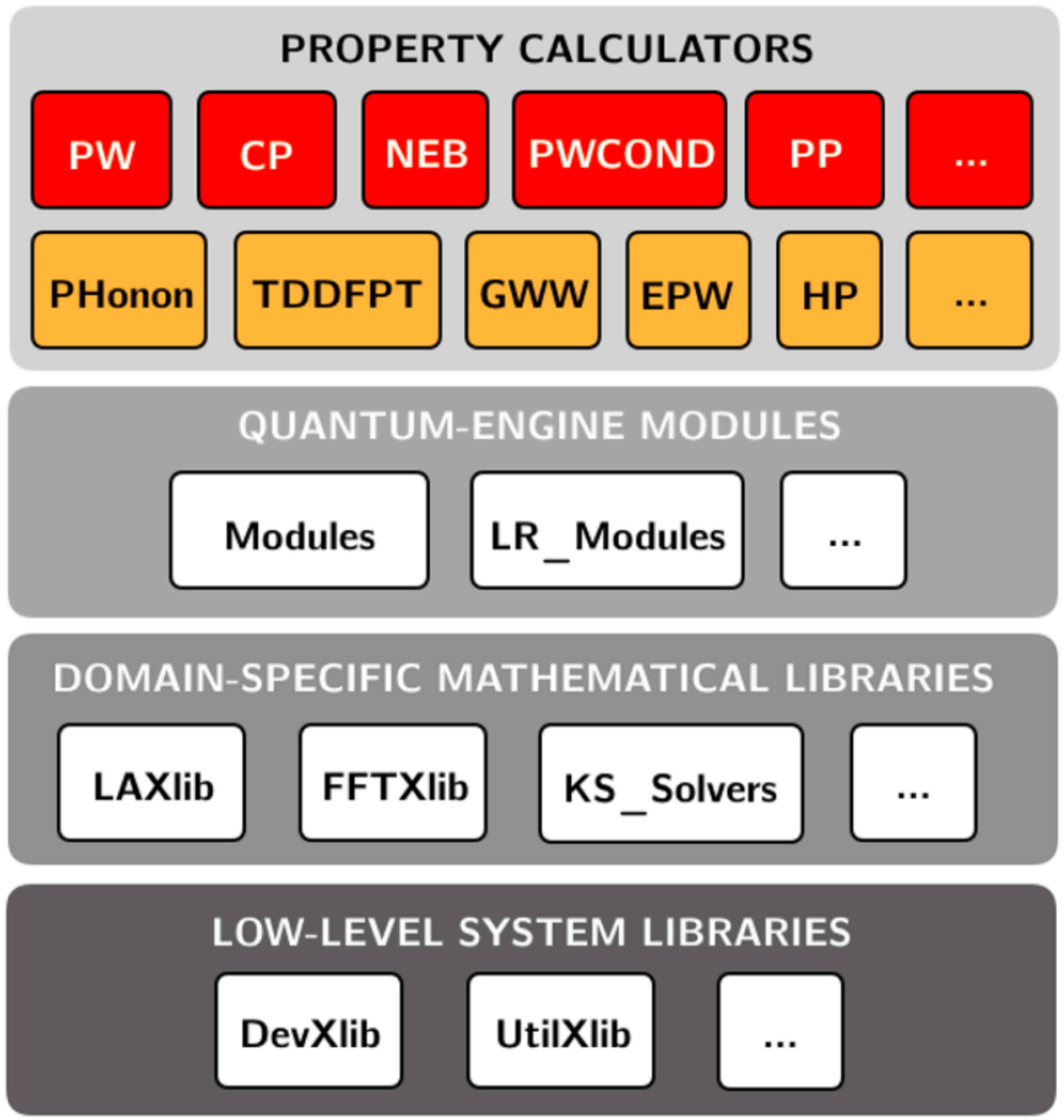
Quantum ESPRESSO



~15 papers cite Quantum ESPRESSO every day

P. Giannozzi, S. Baroni *et al.*, J. Phys.: Condens. Matter 21, 395502 (2009); *ibid.* 29, 465901 (2017).

Quantum ESPRESSO is a suite of codes

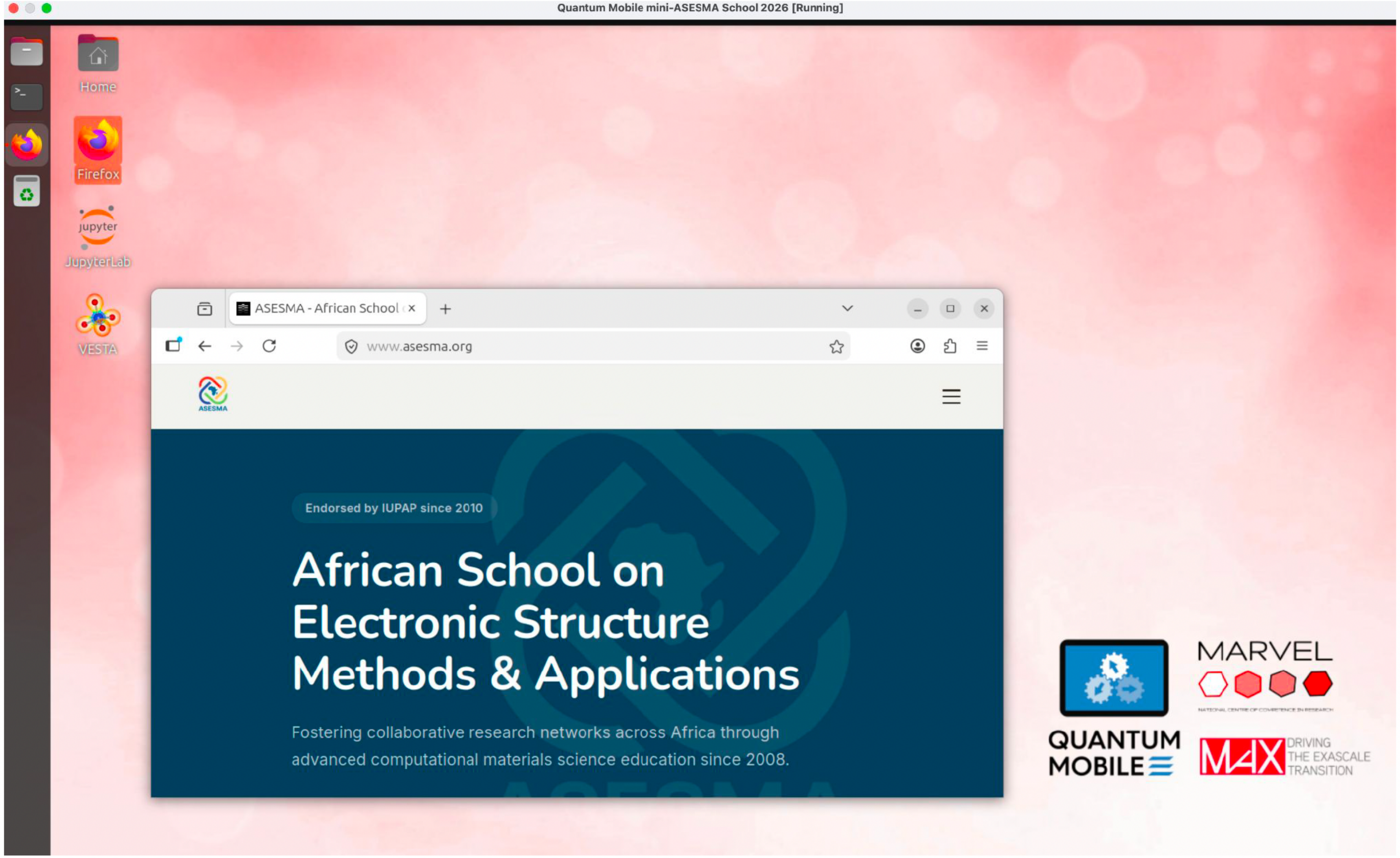


Quantum ESPRESSO is an integrated suite of Open-Source computer codes for electronic-structure calculations and materials modeling at the nanoscale. It is based on density-functional theory, plane waves, and pseudopotentials.

PW - DFT, DFT+U, …
CP - Car-Parrinello molecular dynamics
NEB - Nudged Elastic Bands
PP - Postprocessing tools
Phonon - Phonon calculations
TDDFPT - Spectroscopy
HP - Calculation of Hubbard parameters
…

Quantum Mobile (Plan A)

https://github.com/marvel-nccr/quantum-mobile/blob/support/asesma/custom/README.md



Docker (Plan B)

To start the Docker container for this school, type in the terminal:

asesma_qe

What is Docker?

Docker is a tool for packaging and running applications in isolated environments called “containers”

Why use Docker?

It makes software portable, reproducible, and easy to share

Ideal for schools:

Perfect for scientific codes like Quantum ESPRESSO: install once, run anywhere

In practice:

One command sets up a full working environment for simulations, tutorials, or workshops.

Exercise 1

DFT+*U* for CoO

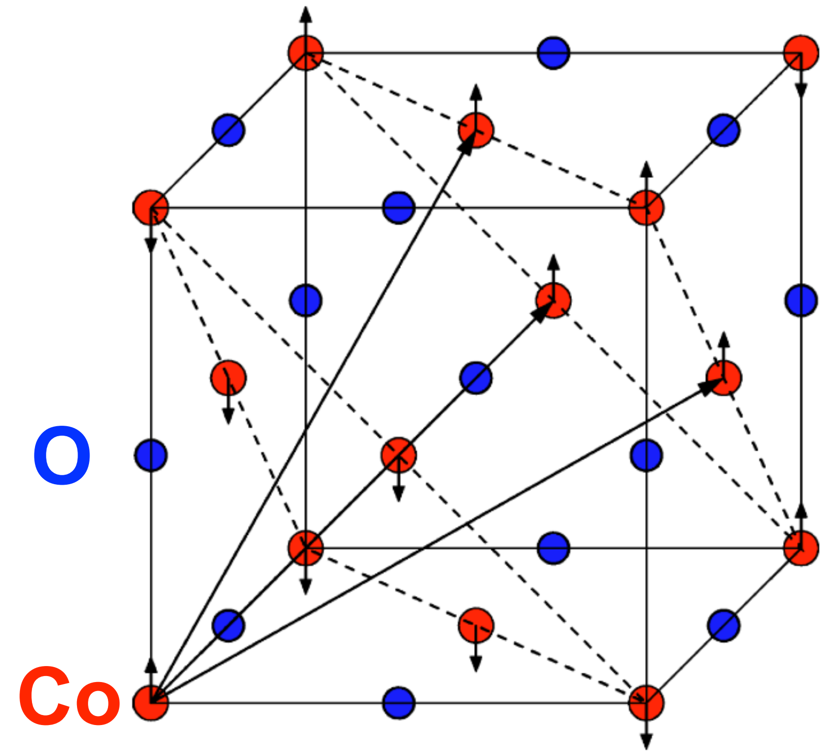
Check README.md how to run the calculations

```
mpirun -n 2 pw.x < CoO.scf.in |tee CoO.scf.out

mpirun -n 2 pw.x < CoO.nscf.in |tee CoO.nscf.out

mpirun -n 2 projwfc.x < CoO.projwfc.in |tee CoO.projwfc.out

python3 plot_pdos.py

Visualize the file CoO_PDOS.pdf
```

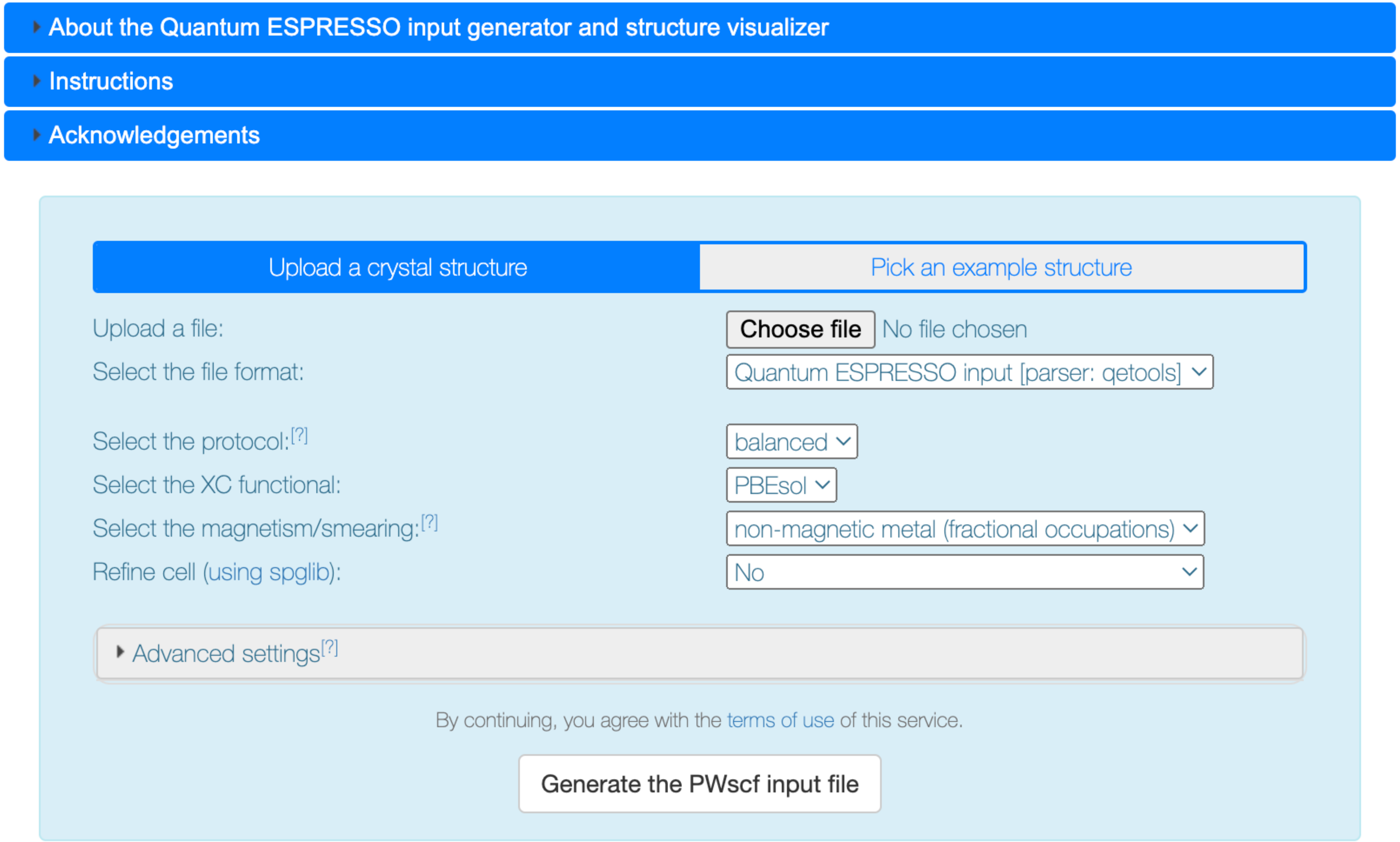
Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
```

- Self-consistent-field (SCF) calculation
- Specification of the lattice (see CELL_PARAMETERS)
- Number of atoms and atomic types
- Kinetic-energy cutoff for the wavefunctions and density/potential
- Spin-polarized collinear calculation (AFM)
- Marzari-Vanderbilt smearing with a broadening parameter of 0.02 Ry
- Convergence threshold for self-consistency
- Pseudopotentials and functional (PBEsol)
- Cell parameters optimized using DFT with the PBEsol functional



Quantum ESPRESSO input generator and structure visualizer



https://www.materialscloud.org/work/tools/qeinputgenerator

SSSP pseudopotential library

- pslibrary.1.0.0 US (high acc.)
- pslibrary.1.0.0 PAW (high acc.)
- pslibrary.0.3.1 US (high acc.)
- pslibrary.0.3.1 PAW (high acc.)
- GBRV-1.2 (US)
- GBRV-1.4 (US)
- GBRV-1.5 (US)
- SG15 (NC)
- SG15-1.1 (NC)
- Pseudo Dojo (NC)
- Goedecker
- THEOS
- RE Wentzcovitch (PAW)
- pslibrary.1.0.0 PAW (low acc.)
- Actinides MalteSachs (PAW)

$E_{WFC(RHO)}$

E: element
WFC: suggested wave function cutoff in Ry for SSSP efficiency *
(RHO): suggested charge density cutoff in Ry for SSSP efficiency (this is equal to the dual times the wave function cutoff)
*the background color corresponds to the pseudopotential library chosen for SSSP efficiency

SSSP Efficiency (version 1.3.0)

Note: The Ir (Iridium) pseudopotential of the efficiency library has a ghost state at ~10 eV above the Fermi energy, please use it with caution.

Cutoffs table | Pseudos | Switch to SSSP Precision

H 60 (480)																	He 50 (200)
Li 40 (320)	Be 40 (320)											B 35 (280)	C 45 (360)	N 60 (480)	O 50 (400)	F 45 (360)	Ne 50 (200)
Na 40 (320)	Mg 30 (240)											Al 30 (240)	Si 30 (240)	P 30 (240)	S 35 (280)	Cl 40 (320)	Ar 60 (240)
K 60 (480)	Ca 30 (240)	Sc 40 (160)	Ti 35 (280)	V 35 (280)	Cr 40 (320)	Mn 65 (780)	Fe 90 (1080)	Co 45 (360)	Ni 45 (360)	Cu 90 (720)	Zn 40 (320)	Ga 70 (560)	Ge 40 (320)	As 35 (280)	Se 30 (240)	Br 30 (240)	Kr 45 (180)
Rb 30 (120)	Sr 30 (240)	Y 35 (280)	Zr 30 (240)	Nb 40 (320)	Mo 35 (140)	Tc 30 (120)	Ru 35 (140)	Rh 35 (140)	Pd 45 (180)	Ag 50 (200)	Cd 60 (480)	In 50 (400)	Sn 60 (480)	Sb 40 (320)	Te 30 (240)	I 35 (280)	Xe 60 (240)
Cs 30 (240)	Ba 30 (240)	*	Hf 50 (200)	Ta 45 (360)	W 30 (240)	Re 30 (240)	Os 40 (320)	Ir 55 (440)	Pt 35 (280)	Au 45 (180)	Hg 50 (200)	Tl 50 (400)	Pb 40 (320)	Bi 45 (360)	Po 75 (600)	At 80 (640)	Rn 120 (960)
Fr 150 (1200)	Ra 90 (720)	**															
		*	La 40 (320)	Ce 50 (400)	Pr 40 (320)	Nd 40 (320)	Pm 40 (320)	Sm 40 (320)	Eu 40 (320)	Gd 40 (320)	Tb 40 (320)	Dy 40 (320)	Ho 40 (320)	Er 40 (320)	Tm 40 (320)	Yb 40 (320)	Lu 45 (360)
		**	Ac 40 (320)	Th 60 (480)	Pa 60 (480)	U 60 (480)	Np 60 (480)	Pu 60 (480)	Am 60 (480)	Cm 60 (480)	Bk 60 (480)	Cf 60 (480)	Es 60 (480)	Fm 60 (480)	Md 60 (480)	No 60 (480)	Lr 60 (480)

https://www.materialscloud.org/discover/sssp/table/efficiency

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
```

Input file CoO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 6 6 6 0 0 0
```

Input file CoO.projwfc.in

```
&projwfc
    prefix='CoO'
    outdir='./tmp'
    ngauss = 0,
    degauss = 0.005,
    Emin = -15.0,
    Emax =  30.0,
    DeltaE = 0.01
 /
```

Gaussian broadening for PDOS

Value of the Gaussian broadening (in Ry)

Minimum and maximum energy for the plot (in eV)

Energy grid step

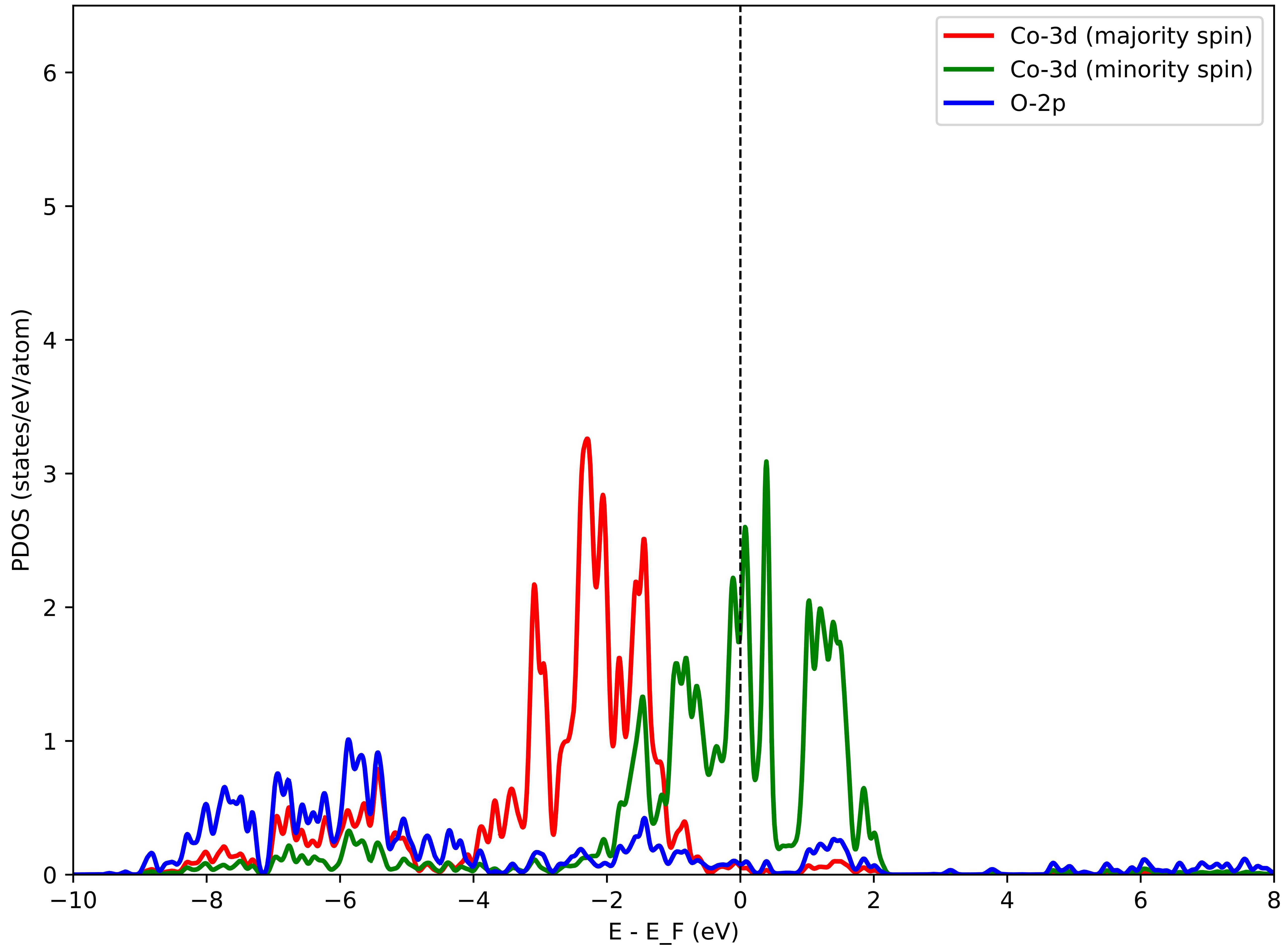
Python3 script: plot_pdos.py

Inspect the script. It aims at plotting Co-3d states (majority spin and minority spin) and O-2p states.

PDOS is shifted such that the Fermi energy corresponds to zero of energy. Use the Fermi energy from the file CoO.scf.out

After running the script, visualize the file CoO_PDOS.pdf

PDOS using DFT (PBEsol)



DFT predicts CoO to be metallic (this is wrong)

Experimentally CoO is insulating

Let's try DFT+*U*

Input file CoO.scf.in

```
 &control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='CoO'
    pseudo_dir = '../../pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 0.0001
U Co2-3d 0.0001
```

HUBBARD card

Name of the card

Type of the Hubbard projectors

```
HUBBARD {ortho-atomic}
U Co1-3d 0.0001
U Co2-3d 0.0001
```

Value of the Hubbard parameter (in eV)

Name of the Hubbard parameter

Hubbard manifold to which we want to apply the Hubbard correction

Atomic type of the chemical element to which we want to apply the Hubbard correction

Detailed description of the new Hubbard input syntax (since v7.1):
See Hubbard_input.pdf

We set some small initial value
just to activate the Hubbard machienary in the code

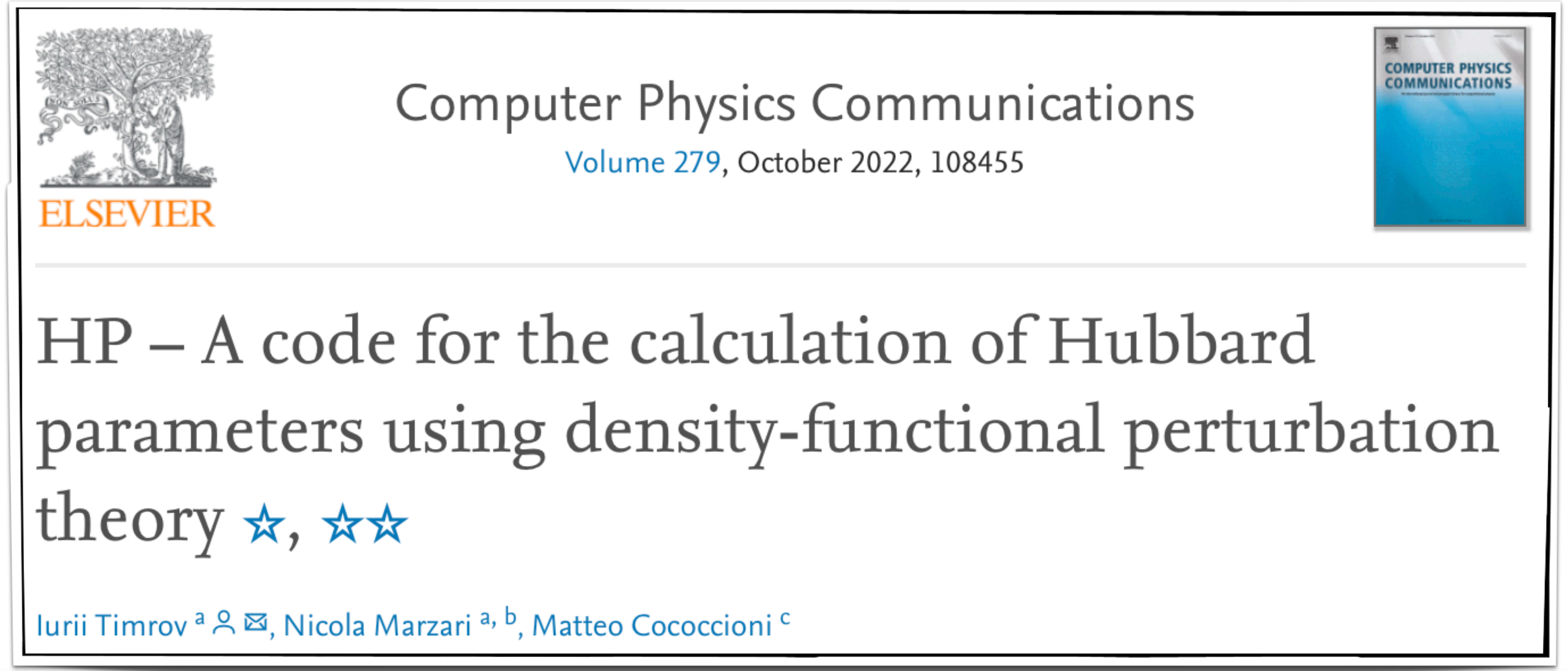
Input file CoO.hp.in

```
&inputhp
    prefix = 'CoO'
    outdir = './tmp'
    nq1 = 2, nq2 = 2, nq3 = 2
    conv_thr_chi = 1.d-6
/
```

The same as in the CoO.scf.in input

Size of the **q** points mesh

Convergence threshold for computing the self-consistent response matrix χ (in eV^{-1})


ELSEVIER

Computer Physics Communications

Volume 279, October 2022, 108455

HP – A code for the calculation of Hubbard parameters using density-functional perturbation theory ☆, ☆☆

Iurii Timrov [a], Nicola Marzari [a, b], Matteo Cococcioni [c]

Important notice: The calculation of the Hubbard *U* parameter must be converged (as any other quantity of interest)!

Hubbard *U* must be converged with respect to the kinetic-energy cutoff (ecutwfc and ecutrho), **k** points mesh, and **q** points mesh.

For more details see: *I. Timrov, N. Marzari, M. Cococcioni, Phys. Rev. B* ***98****, 085127 (2018).*

Output file CoO.Hubbard_parameters.dat

```
=------------------------------------------------------------------=

                     Hubbard U parameters:

   site n.   type   label   spin   new_type   new_label   Hubbard U (eV)
     1         1     Co1      1       1          Co1          6.7553
     2         2     Co2     -1       1          Co1          6.7553

=------------------------------------------------------------------=
```

These are the output Hubbard *U* parameters for Co1-3d and Co2-3d states

These parameters are computed in a "one-shot" fashion (i.e. from the DFT-PBEsol ground state)

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

HUBBARD card


```
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```


This is the value of the Hubbard *U* parameter (in eV) that we computed using the HP code

We need to specify the same Hubbard *U* in the CoO.nscf.in file

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

Input file CoO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

Input file CoO.projwfc.in

This is exactly the same as before

```
&projwfc
    prefix='CoO'
    outdir='./tmp'
    ngauss = 0,
    degauss = 0.005,
    Emin = -15.0,
    Emax =  30.0,
    DeltaE = 0.01
 /
```

- ngauss: Gaussian broadening for PDOS
- degauss: Value of the Gaussian broadening (in Ry)
- Emin, Emax: Minimum and maximum energy for the plot (in eV)
- DeltaE: Energy grid step

Python3 script: plot_pdos.py

Plot the PDOS using the gnuplot and the plot_pdos.py script

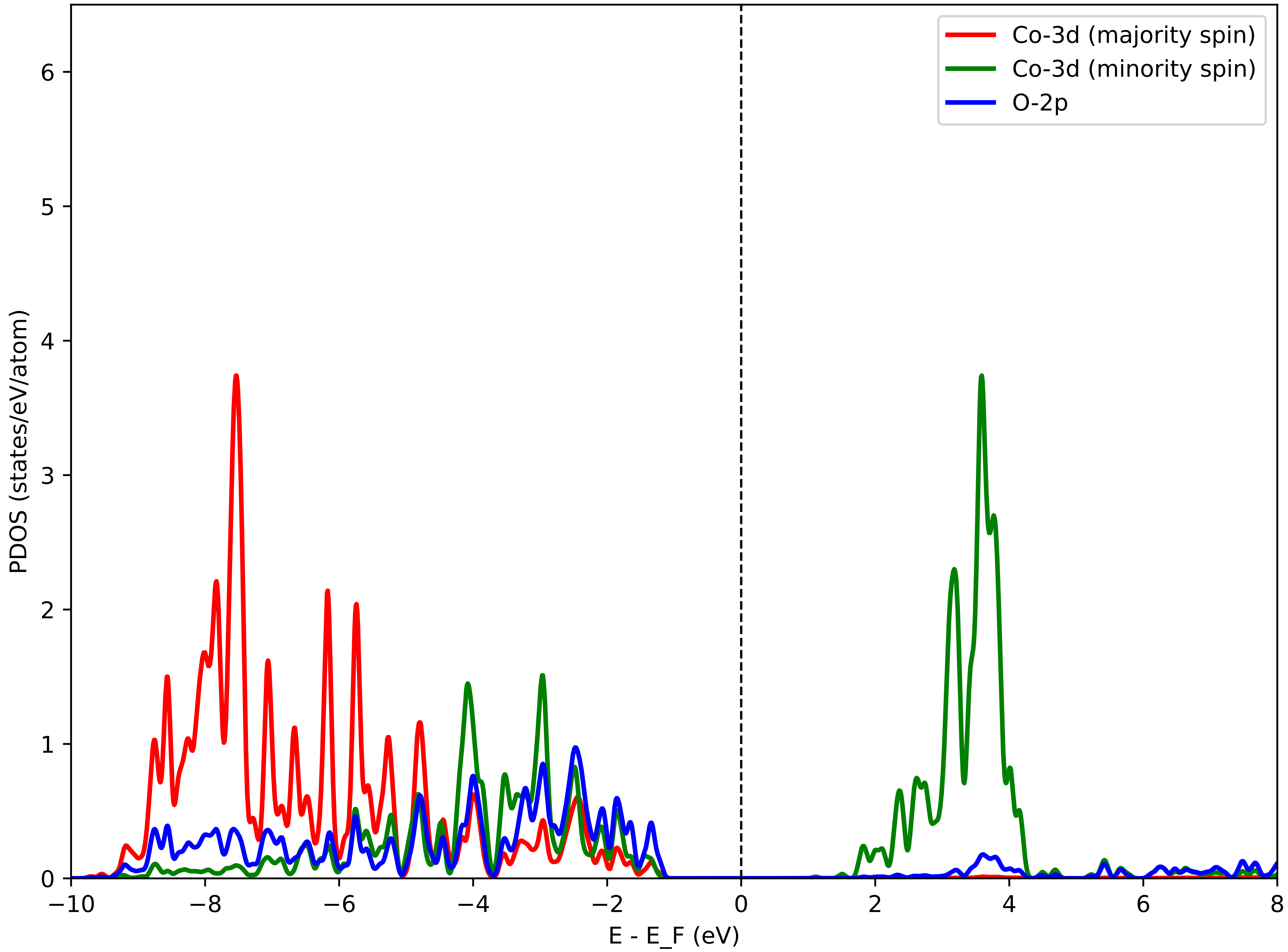
PDOS is shifted such that the Fermi energy corresponds to zero of energy. Use the Fermi energy from the file CoO.scf.out

After running the script, visualize the file CoO_PDOS.pdf

Run the calculations

```
mpirun -n 2 pw.x < CoO.scf.in |tee CoO.scf.out

mpirun -n 2 pw.x < CoO.nscf.in |tee CoO.nscf.out

mpirun -n 2 projwfc.x < CoO.projwfc.in |tee CoO.projwfc.out

python3 plot_pdos.py

Visualize the file CoO_PDOS.pdf
```

PDOS using DFT+*U*

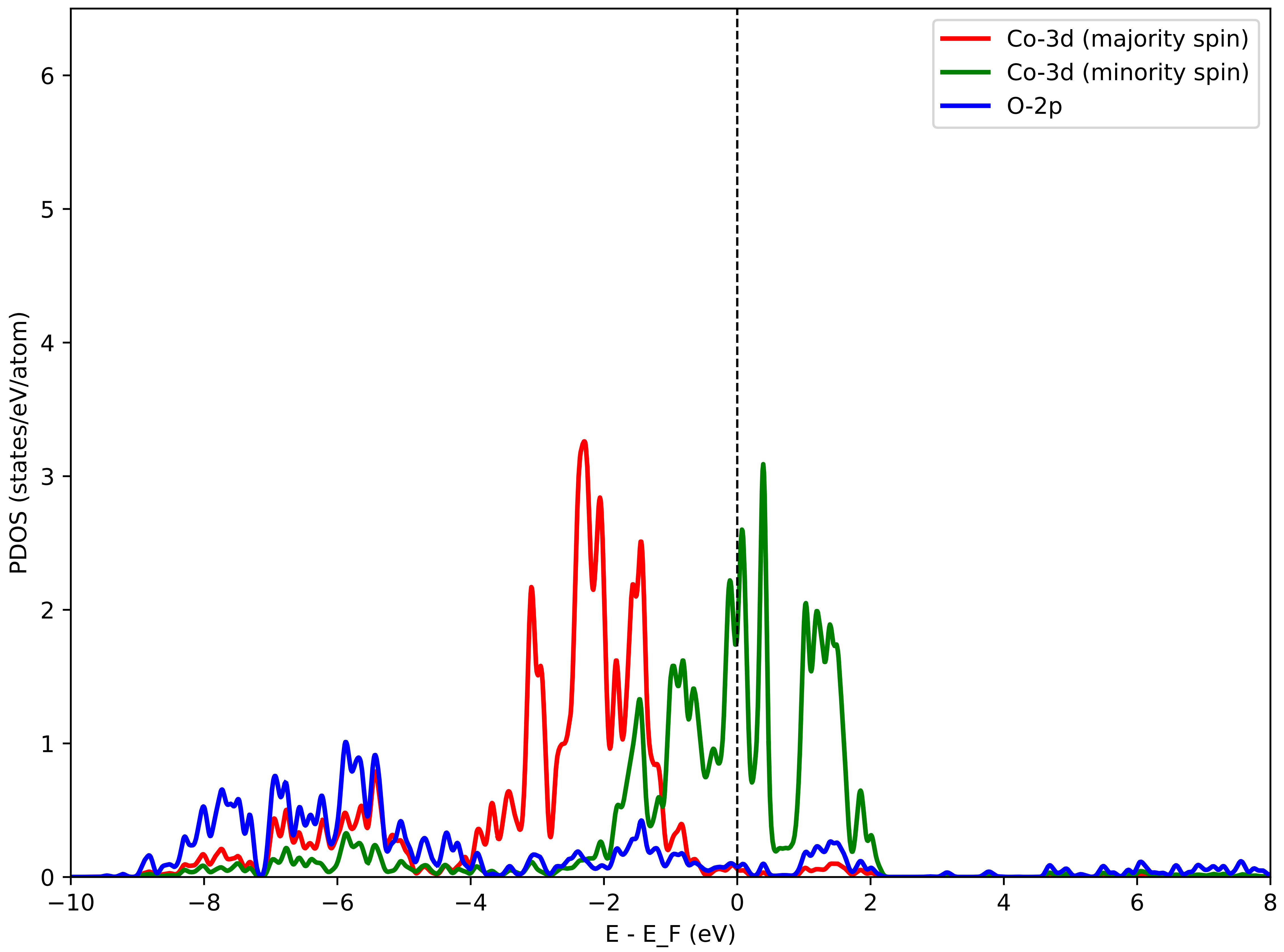


DFT+*U* predicts CoO to be insulating (this is correct)

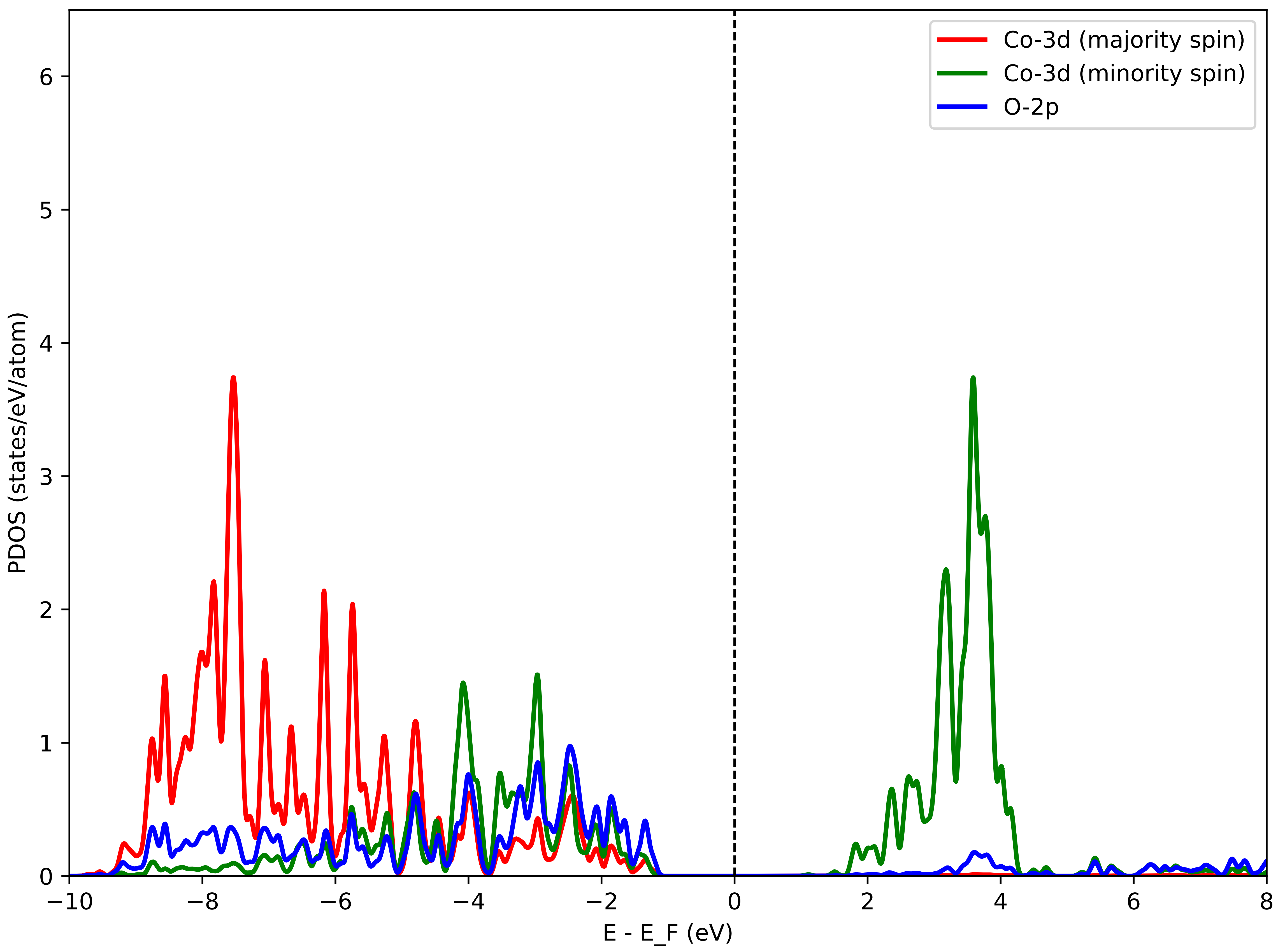
Experimentally CoO is insulating

Agreement with experiment

PDOS using DFT



PDOS using DFT+*U*



Co-3d (minority spin) states are split

Co-3d (majority spin) states are shifted to lower energies

DFT+*U* band gap: 2.4 eV Experimental gap: 2.5±0.3 eV

Exercise 2

Hybrid functional PBE0 for Si

The $\mathbf{q} + \mathbf{G} = 0$ divergence

In periodic systems, the exact exchange contains a divergence when $\mathbf{q} + \mathbf{G} = 0$:

$$E_x = -\frac{4\pi}{2\Omega} \times \frac{\Omega}{(2\pi)^3} \int d\mathbf{q} \sum_{\mathbf{G}} \frac{A(\mathbf{q}+\mathbf{G})}{|\mathbf{q}+\mathbf{G}|^2}$$

We need to setup the $\mathbf{q}$ mesh

where

We need to setup the $\mathbf{k}$ mesh

$$A(\mathbf{q}+\mathbf{G}) = \frac{\Omega}{(2\pi)^3} \int d\mathbf{k} |\rho_{\mathbf{k},v}^{\mathbf{k}-\mathbf{q},v'}(\mathbf{q}+\mathbf{G})|^2 \equiv \frac{1}{N_k} \sum_{\mathbf{k}} |\rho_{\mathbf{k},v}^{\mathbf{k}-\mathbf{q},v'}(\mathbf{q}+\mathbf{G})|^2$$

(the finite sum over N_k **k**-points is what we actually compute) and

$$\rho_{\mathbf{k},v}^{\mathbf{k}-\mathbf{q},v'}(\mathbf{r}) = \psi^*_{\mathbf{k}-\mathbf{q},v'}(\mathbf{r}) \psi_{\mathbf{k},v}(\mathbf{r}).$$

This divergence is integrable, see: F.Gygi and A.Baldereschi, PRB 34, 4405 (1986)

Input file Si.scf.in

```
&control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='Si'
    pseudo_dir = '../../pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 2,
    celldm(1) = 10.20,
    nat = 2,
    ntyp = 1,
    ecutwfc = 20.0,
    input_dft = 'pbe0',
    nqx1 = 1, nqx2 = 1, nqx3 = 1,
    x_gamma_extrapolation = .true.
    exxdiv_treatment = 'gygi-baldereschi'
 /
 &electrons
    conv_thr =  1.d-9
    mixing_beta = 0.3
 /
ATOMIC_SPECIES
 Si  28.086  Si.pbe-rrkj.UPF
ATOMIC_POSITIONS {alat}
 Si 0.00 0.00 0.00
 Si 0.25 0.25 0.25
K_POINTS {automatic}
 8 8 8 1 1 1
```

Type of the hybrid functional (PBE0 in this case)

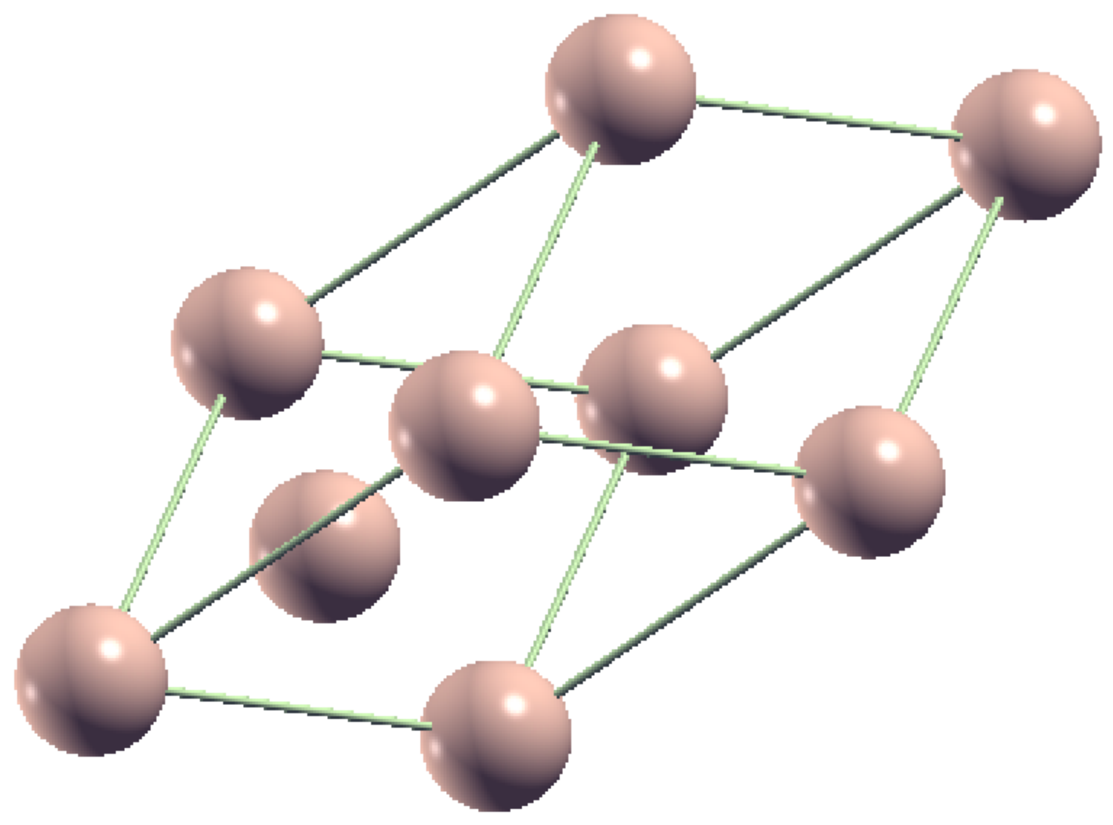
Override the functional written in the pseudopotential (PP) file.
Important: use a PP for the closest GGA (PBE in this case), there are no PP's for hybrids.

q-point mesh

1x1x1 is the minimal mesh (**q**=0), it is fast but not accurate

Treatment of the $\mathbf{q}+\mathbf{G}=0$ divergence

Singularity is analytically integrated



Popular hybrid functionals

input_dft = “PBE0”

J.P.Perdew, M. Ernzerhof, K.Burke, JCP 105, 9982 (1996)

C. Adamo, V. Barone, JCP 110, 6158 (1999)

input_dft = “B3LYP”

P.J. Stephens,F.J. Devlin,C.F. Chabalowski,M.J. Frisch, J.Phys.Chem 98, 11623 (1994)

input_dft = “HSE”

Heyd, Scuseria, Ernzerhof, J. Chem. Phys. 118, 8207 (2003)

Heyd, Scuseria, Ernzerhof, J. Chem. Phys. 124, 219906 (2006)

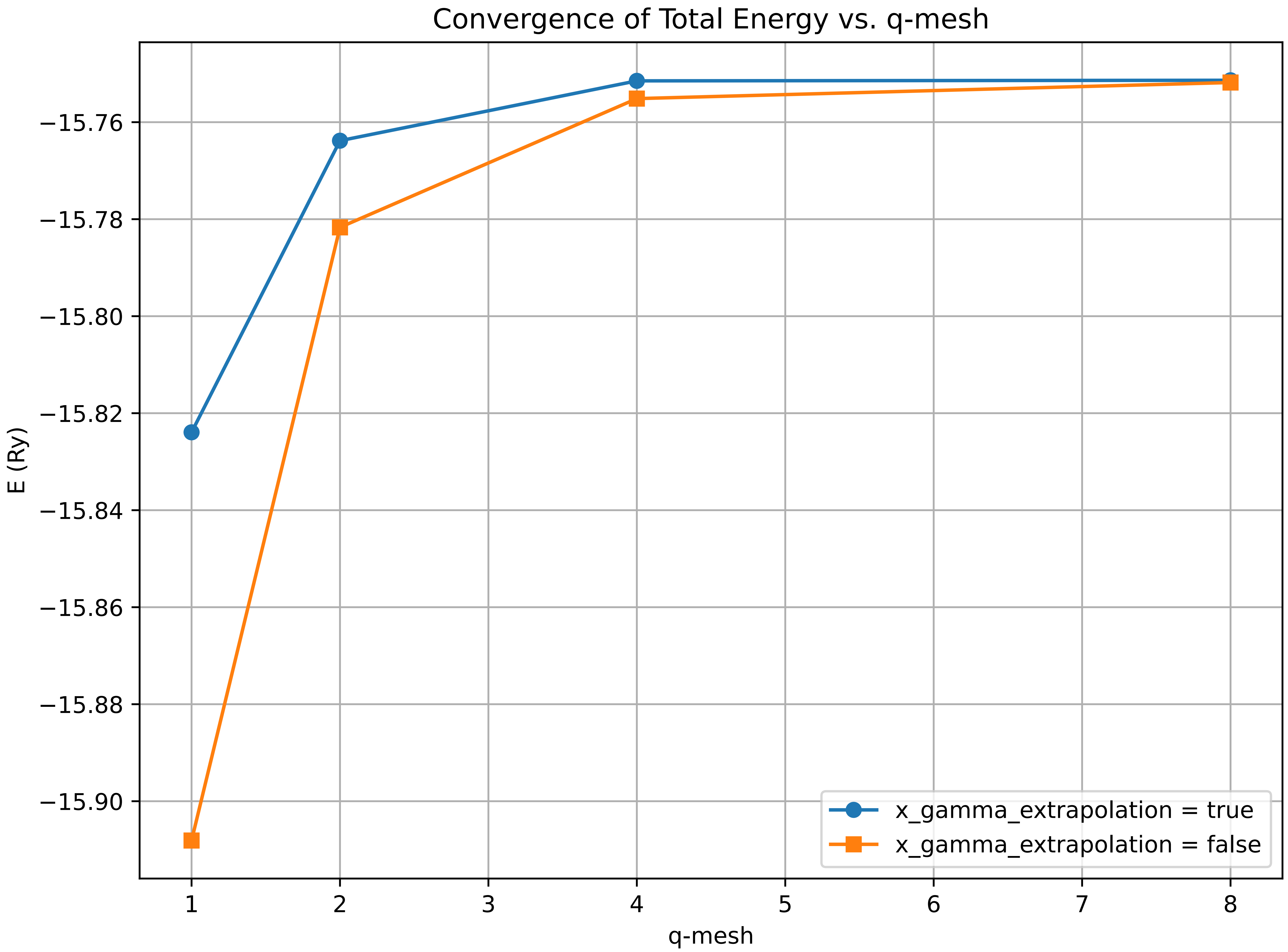
Types of treatment of the $\mathbf{q} + \mathbf{G} = 0$ divergence

exxdiv_treatment	CHARACTER
Default:	'gygi-baldereschi'

```
Specific for EXX. It selects the kind of approach to be used
for treating the Coulomb potential divergencies at small q vectors.

   'gygi-baldereschi' :
        appropriate for cubic and quasi-cubic supercells

   'vcut_spherical' :
        appropriate for cubic and quasi-cubic supercells

   'vcut_ws' :
        appropriate for strongly anisotropic supercells, see also ecutvcut.

   'none' :
        sets Coulomb potential at G,q=0 to 0.0 (required for GAU-PBE)
```

Check the online documentation at https://www.quantum-espresso.org/Doc/INPUT_PW.html

Convergence of the total energy with respect to the $\mathbf{q}$-point mesh

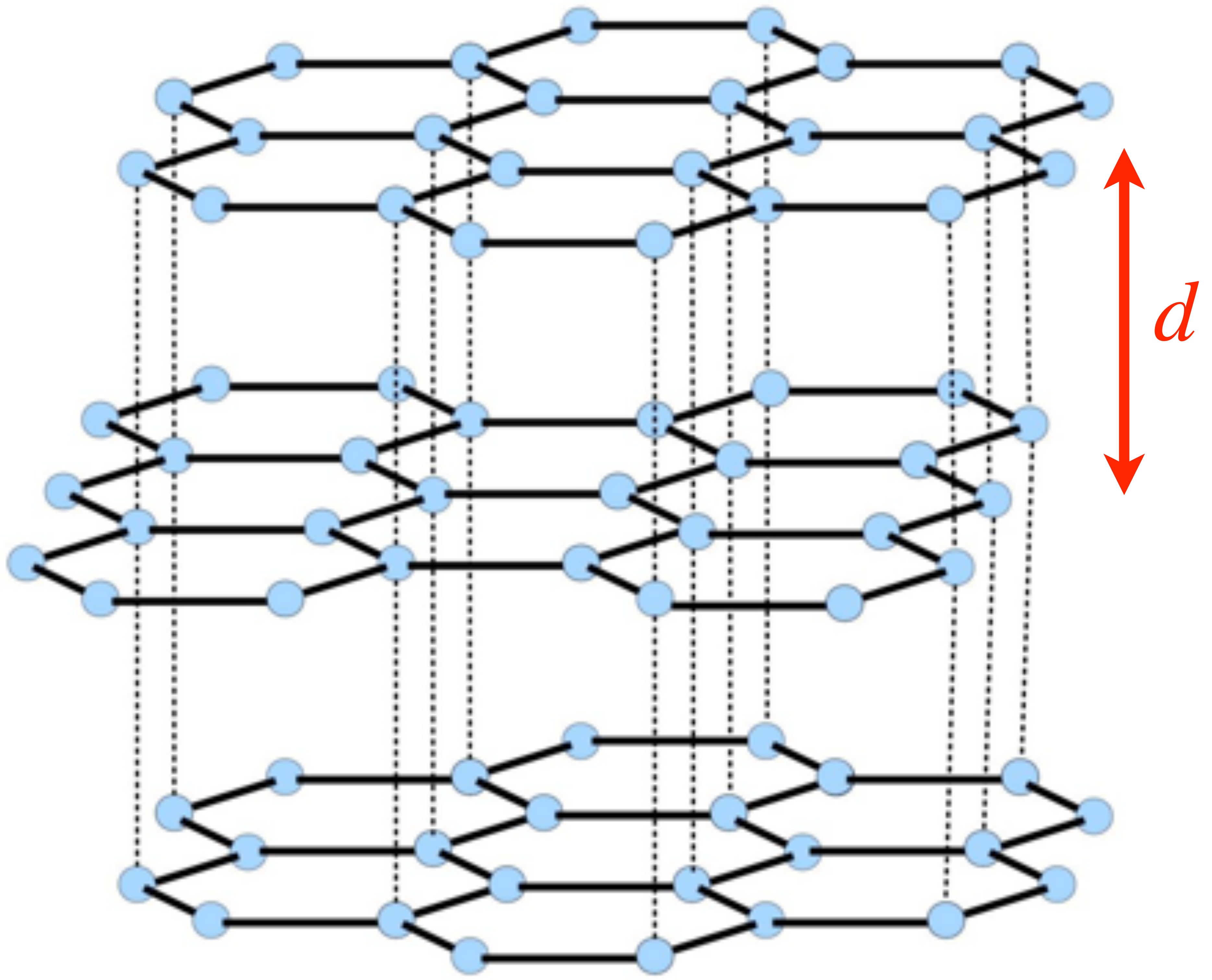


Convergence is faster when x_gamma_extrapolation = .true.

Exercise 3

Van der Waals functionals for graphite

Graphite



There are van der Waals (dispersive) interactions between layers in graphite.

The equilibrium inter-layer distance d is too small within LDA and too large within GGA with respect to the experimental value (3.336 A).

Van der Waals interactions must be taken into account!

Input file graphite.vc-relax.in

```
&control
    calculation='vc-relax'
    restart_mode='from_scratch',
    prefix='graphite'
    pseudo_dir = '../../pseudo'
    outdir='./tmp/'
    verbosity='high'
    etot_conv_thr = 1.0D-5
    forc_conv_thr = 1.0D-4
/
&system
   ibrav = 4,
   a = 2.466,
   c = 6.411,
   nat = 4,
   ntyp = 1,
   ecutwfc = 30.0,
   ecutrho = 240.0,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02,
   input_dft = 'vdw-DF'
/
&electrons
   conv_thr =  1.d-9
   mixing_beta = 0.3
/
&ions
/
&cell
/
ATOMIC_SPECIES
 C   12.011    C.pbe-rrkjus.UPF
ATOMIC_POSITIONS {crystal}
 C      0.000000    1.000000    0.75000
 C      0.666667    0.333333    0.75000
 C      0.000000    1.000000    0.25000
 C      0.333333    0.666667    0.25000
K_POINTS {automatic}
 4 4 2 1 1 1
```

Variable-cell optimization

Hexagonal lattice, a and c are in Angtrom!

Type of the van Der Waals functional (vdw-DF in this case)

Override the functional written in the PP file.
Important: use PP for the closest GGA (PBE in this case), because there are no PP for non-local functionals

The **k** point mesh is denser along X and Y than along Z (in reciprocal space), reflecting shorter periodicity in the XY plane

Types of the Van der Waals corrections

vdw_corr	CHARACTER
Default:	'none'
See:	london_s6, london_rcut, london_c6, london_rvdw, dftd3_version, dftd3_threebody, ts_vdw_econv_thr, ts_vdw_isolated, xdm_a1, xdm_a2

```
Type of Van der Waals correction. Allowed values:

   'grimme-d2', 'Grimme-D2', 'DFT-D', 'dft-d' :
        Semiempirical Grimme's DFT-D2. Optional variables:
        london_s6, london_rcut, london_c6, london_rvdw
        S. Grimme, J. Comp. Chem. 27, 1787 (2006), doi:10.1002/jcc.20495
        V. Barone et al., J. Comp. Chem. 30, 934 (2009), doi:10.1002/jcc.21112

   'grimme-d3', 'Grimme-D3', 'DFT-D3', 'dft-d3' :
        Semiempirical Grimme's DFT-D3. Optional variables:
        dftd3_version, dftd3_threebody
        S. Grimme et al, J. Chem. Phys 132, 154104 (2010), doi:10.1063/1.3382344

   'TS', 'ts', 'ts-vdw', 'ts-vdW', 'tkatchenko-scheffler' :
        Tkatchenko-Scheffler dispersion corrections with first-principle derived
        C6 coefficients.
        Optional variables: ts_vdw_econv_thr, ts_vdw_isolated
        See A. Tkatchenko and M. Scheffler, PRL 102, 073005 (2009).

   'MBD', 'mbd', 'many-body-dispersion', 'mbd_vdw' :
        Many-body dipersion (MBD) correction to long-range interactions.
        Optional variables: ts_vdw_isolated
        A. Ambrosetti, A. M. Reilly, R. A. DiStasio, A. Tkatchenko, J. Chem. Phys. 140
        18A508 (2014).

   'XDM', 'xdm' :
        Exchange-hole dipole-moment model. Optional variables: xdm_a1, xdm_a2
        A. D. Becke et al., J. Chem. Phys. 127, 154108 (2007), doi:10.1063/1.2795701
        A. Otero de la Roza et al., J. Chem. Phys. 136, 174109 (2012),
        doi:10.1063/1.4705760

 Note that non-local functionals (eg vdw-DF) are NOT specified here but in input_dft
```

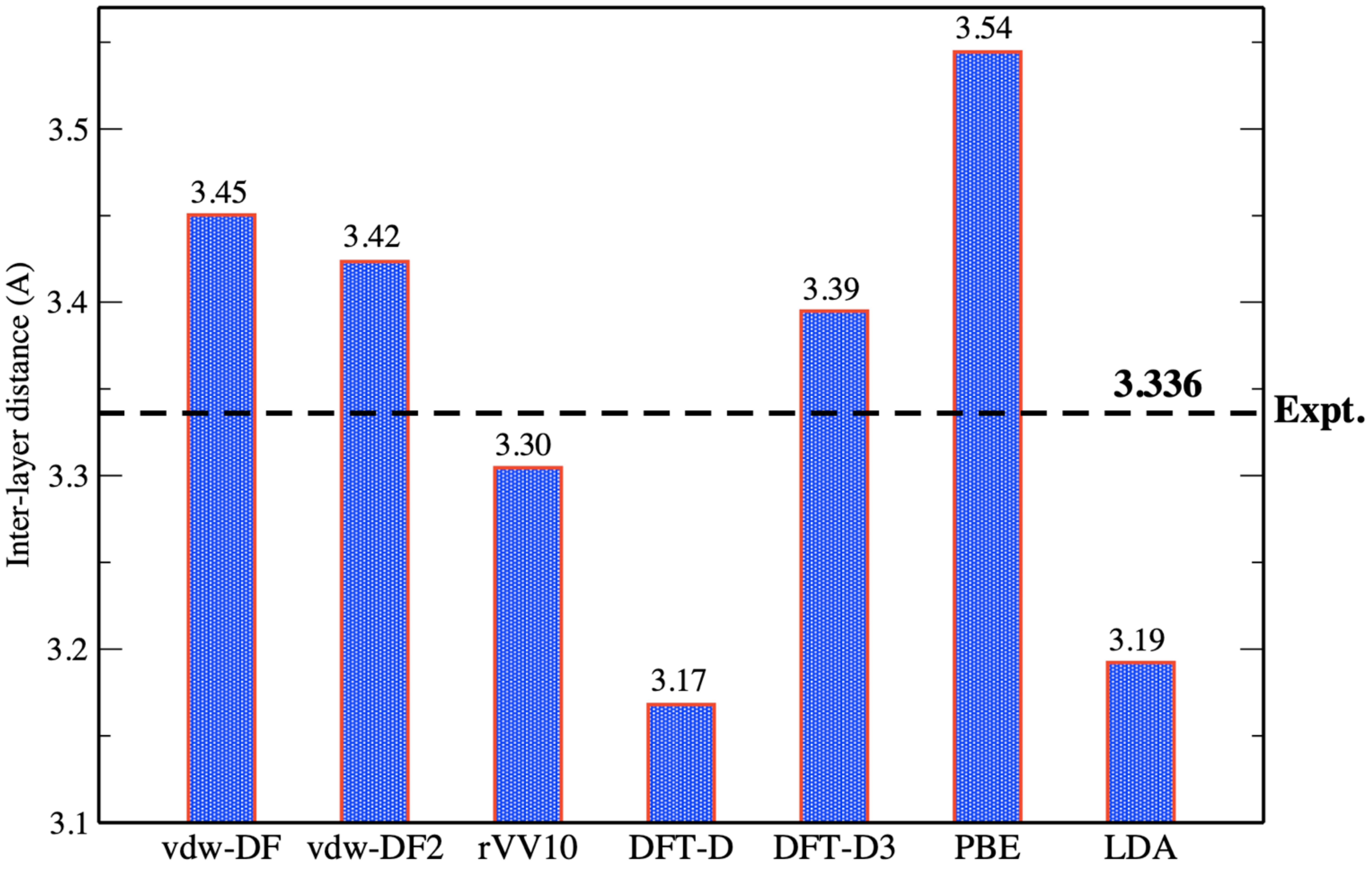
Check the online documentation at https://www.quantum-espresso.org/Doc/INPUT_PW.html

Structural optimization of graphite

```
Study different cases:
 1. input_dft = 'vdw-DF'      @PBE pseudo (non-local)       C.pbe-rrkjus.UPF
 2. input_dft = 'vdw-DF2'     @PBE pseudo (non-local)       C.pbe-rrkjus.UPF
 3. input_dft = 'rVV10'       @PBE pseudo (non-local)       C.pbe-rrkjus.UPF
 4. vdw_corr  = 'DFT-D'       @PBE pseudo (semi-empirical)  C.pbe-rrkjus.UPF
 5. vdw_corr  = 'DFT-D3'      @PBE pseudo (semi-empirical)  C.pbe-rrkjus.UPF
 6. Normal PBE calculation    @PBE pseudo                   C.pbe-rrkjus.UPF
 7. Normal LDA calculation    @LDA pseudo                   C.pz-rrkjus.UPF
```

Use XCrySDen (outside the Docker) to determine the inter-layer distances after the optimization

Inter-layer distance in graphite



Note: these results must be carefully converged (cutoff, k points, etc.)

Exercise 4

meta-GGA for Si and Fe

Input file

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='Si'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 2,
   celldm(1) = 10.262,
   nat = 2,
   ntyp = 1,
   ecutwfc = 40.0,
   input_dft = 'SCAN'
   nbnd = 5
/
&electrons
   conv_thr = 1.d-10
/
ATOMIC_SPECIES
Si 28.086  Si.SCAN.UPF
ATOMIC_POSITIONS {alat}
 Si 0.00 0.00 0.00
 Si 0.25 0.25 0.25
K_POINTS {automatic}
 12 12 12 0 0 0
```

Experimental lattice parameter

It is necessary to perform convergence tests w.r.t ecutwfc

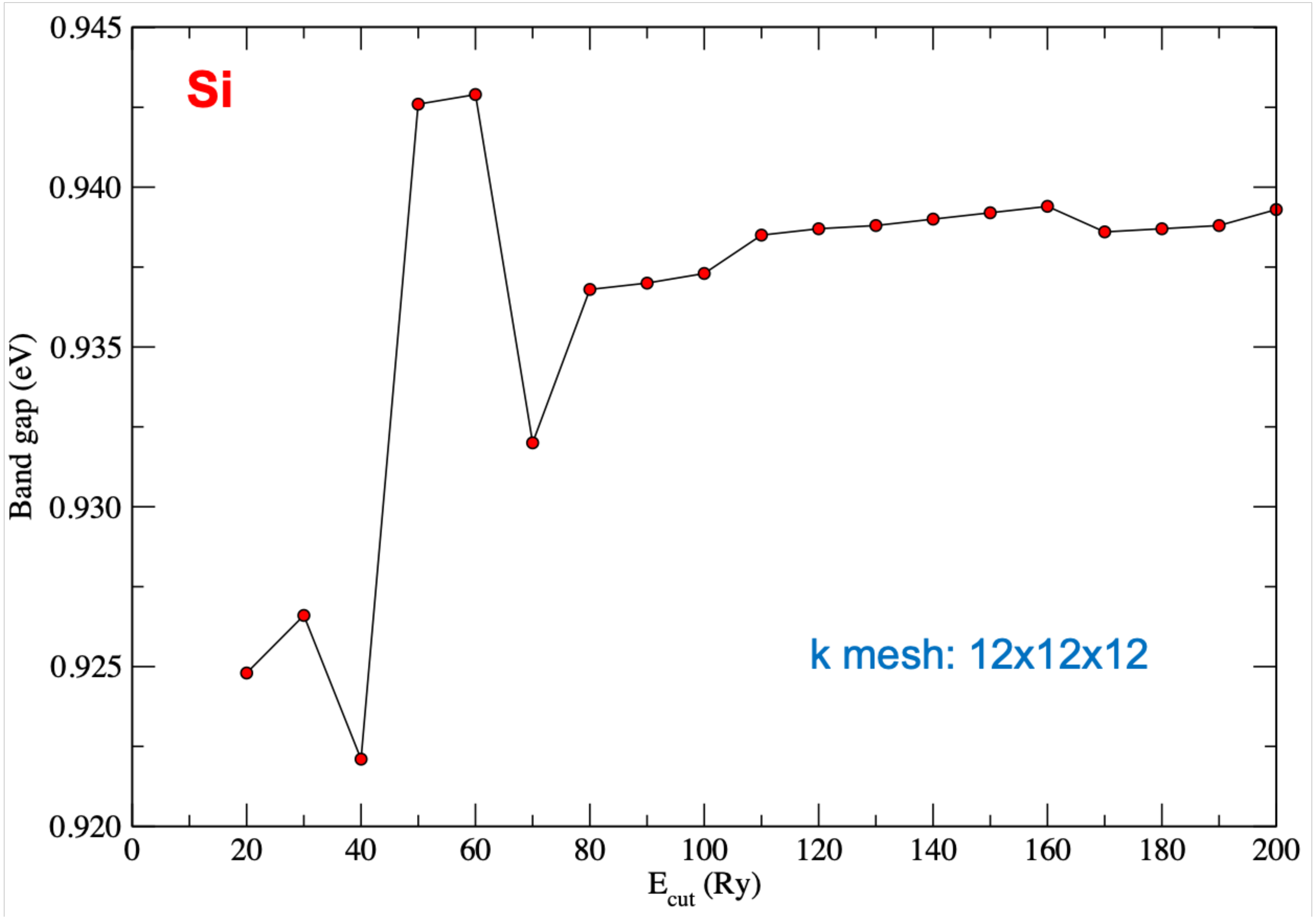
Set up the SCAN functional from the input

Quantum ESPRESSO must be compiled with Libxc

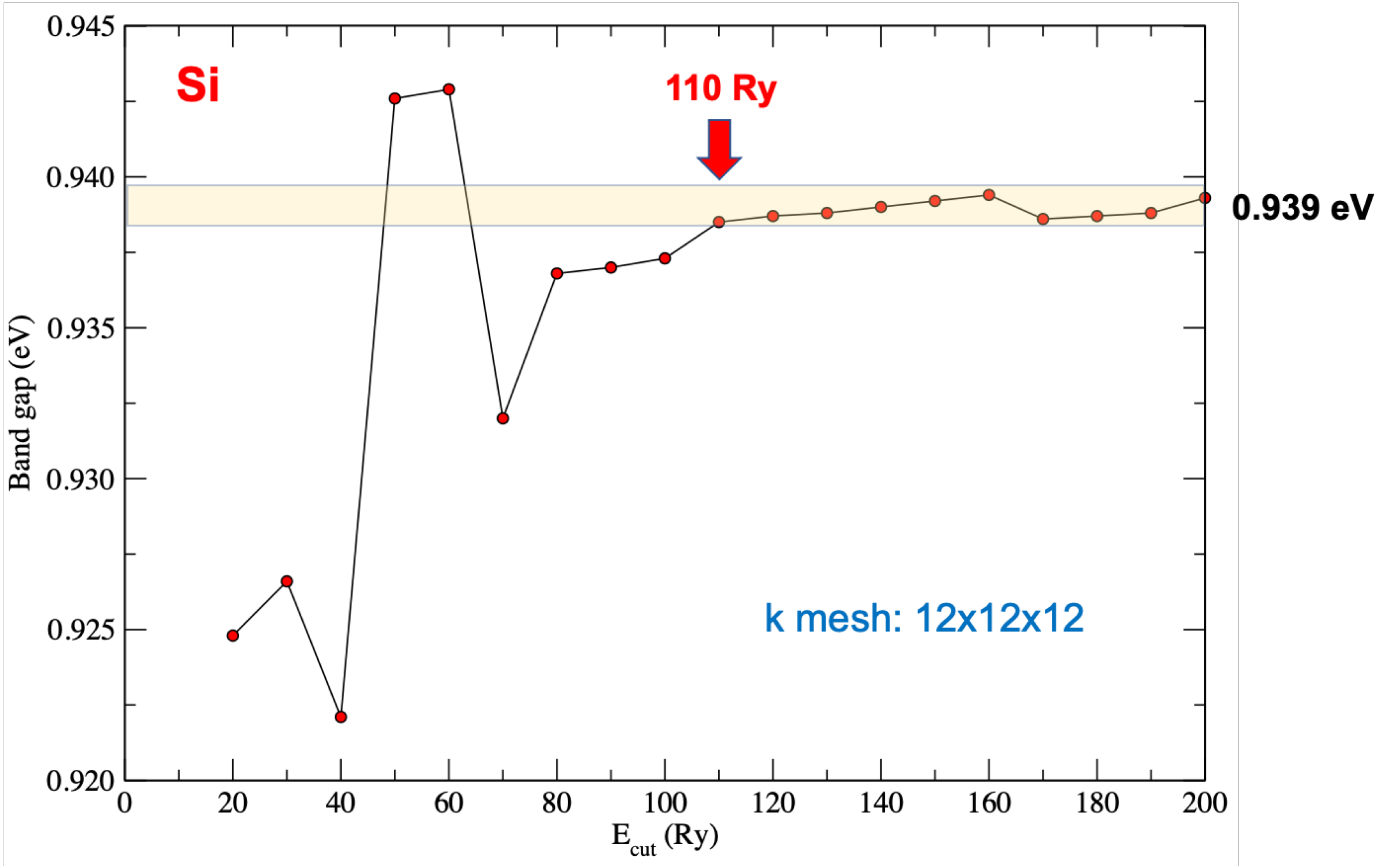
Pseudopotential generated using the SCAN functional

https://yaoyi92.github.io/scan-tm-pseudopotentials.html

SCAN functional & SCAN pseudopotential



SCAN functional & SCAN pseudopotential



SCAN functional & PBE pseudopotential

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='Si'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 2,
   celldm(1) = 10.262,
   nat = 2,
   ntyp = 1,
   ecutwfc = 110.0,
   input_dft = 'SCAN'
   nbnd = 5
/
&electrons
   conv_thr = 1.d-10
/
ATOMIC_SPECIES
Si 28.086  Si.pbe_PseudoDojo.UPF
ATOMIC_POSITIONS {alat}
 Si 0.00 0.00 0.00
 Si 0.25 0.25 0.25
K_POINTS {automatic}
 12 12 12 0 0 0
```

Pseudo Dojo library
http://www.pseudo-dojo.org

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='Si'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 2,
   celldm(1) = 10.262,
   nat = 2,
   ntyp = 1,
   ecutwfc = 110.0,
   input_dft = 'SCAN'
   nbnd = 5
/
&electrons
   conv_thr = 1.d-10
/
ATOMIC_SPECIES
Si 28.086  Si_ONCV_PBE-1.2.upf
ATOMIC_POSITIONS {alat}
 Si 0.00 0.00 0.00
 Si 0.25 0.25 0.25
K_POINTS {automatic}
 12 12 12 0 0 0
```

SG15 library
http://www.quantum-simulation.org/potentials/sg15_oncv/

SCAN functional with different pseudopotentials

Ecut = 110 (Ry), k mesh: 12x12x12

Functional	Pseudopotential	Library	Gap (eV)
SCAN	SCAN	Yi Yao's library	0.94
SCAN	PBE	SG15 ONCV	0.83
SCAN	PBE	Pseudo Dojo	0.64

Gap_{expt} = 1.17 (eV)

SCAN functional with different pseudopotentials

Ecut = 110 (Ry), k mesh: 12x12x12

Functional	Pseudopotential	Library	Gap (eV)
SCAN	SCAN	Yi Yao's library	0.94
SCAN	PBE	SG15 ONCV	0.83
SCAN	PBE	Pseudo Dojo	0.64

NLCC=.false.

NLCC=.false.

NLCC=.true.

Gap_{expt} = 1.17 (eV)

Currently, nonlinear core correction (NLCC) is not implemented for meta-GGA in Quantum ESPRESSO!

Pseudopotentials that have NLCC=.true. introduce some inconsistency for meta-GGA calculations.

SCAN functional with different pseudopotentials

Ecut = 110 (Ry), k mesh: 12x12x12

Functional	Pseudopotential	Library	Gap (eV)
SCAN	SCAN	Yi Yao's library	0.94
SCAN	PBE	SG15 ONCV	0.83
SCAN	PBE	Pseudo Dojo	0.64
PBE	PBE	SG15 ONCV	0.56
PBE	PBE	Pseudo Dojo	0.57

Gap_{expt} = 1.17 (eV)

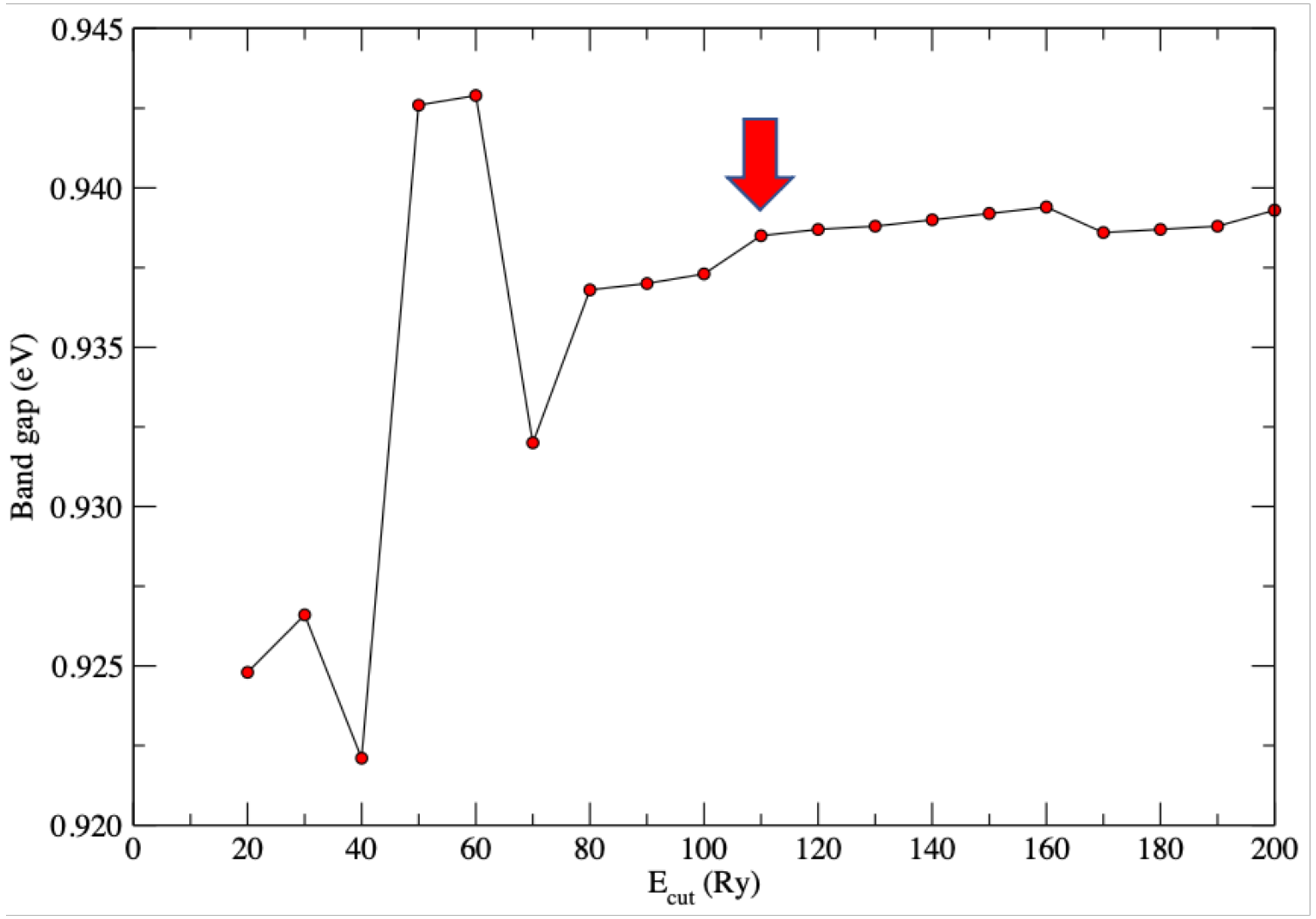
SCAN calculation with the SCAN pseudopotential gives the most accurate results

Y. Yao and Y. Kanai, J. Chem. Phys. ***146****, 224105 (2017).*

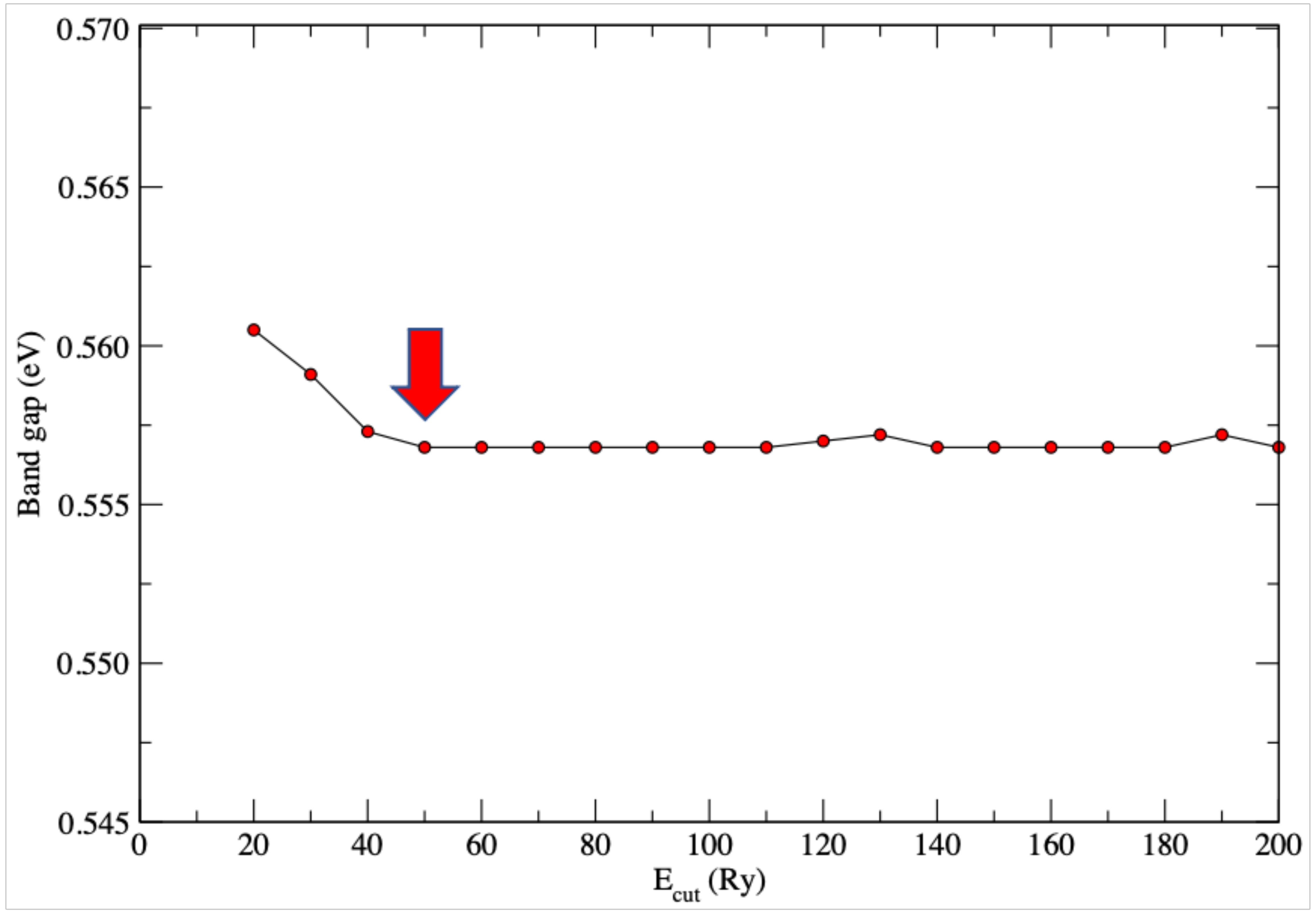
Convergence with respect to E_{cut}

k mesh: 12x12x12

SCAN with SCAN pseudopotential



PBE with PBE pseudopotential (SG15)



Convergence is achieved must faster with PBE than with SCAN

Setting up the XC functional

GGA exchange
GGA correlation

input_dft = 'PBE' or input_dft='XC-000I-000I-101L-130L-000I-000I'

LDA exchange
LDA correlation
meta-GGA exchange
meta-GGA correlation

Setting up the XC functional

GGA exchange GGA correlation

input_dft = 'PBE' or input_dft='XC-000I-000I-101L-130L-000I-000I'

LDA exchange LDA correlation meta-GGA exchange meta-GGA correlation

input_dft = 'SCAN' or input_dft='XC-000I-000I-000I-000I-263L-267L'

no short name in QE ~~input_dft = 'rSCAN'~~ or input_dft='XC-000I-000I-000I-000I-493L-494L'

input_dft = 'r2SCAN' or input_dft='XC-000I-000I-000I-000I-497L-498L'

Read more about this here: https://www.quantum-espresso.org/Doc/user_guide/node13.html

Libxc with IDs: https://libxc.gitlab.io/

Different flavors of SCAN

Functional	Pseudopotential	Library	Gap (eV)
SCAN	PBE	SG15 ONCV	0.83
rSCAN	PBE	SG15 ONCV	0.71
r^2SCAN	PBE	SG15 ONCV	0.70

Gap_{expt} = 1.17 (eV)

SCAN is more accurate than rSCAN and r^2SCAN for predicting the band gap of bulk Si
(at least when using the PBE pseudopotential)

Input file

```
 &control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='Fe'
    pseudo_dir = '../../pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 3,
    celldm(1) = 5.418,
    nat = 1,
    ntyp = 1,
    ecutwfc = 50.0,
    occupations = 'smearing',
    smearing ='mv',
    degauss = 0.02,
    nspin = 2,
    starting_magnetization(1) = 0.5,
    input_dft = 'SCAN'
 /
 &electrons
    conv_thr = 1.d-10
 /
ATOMIC_SPECIES
Fe  55.845  Fe_ONCV_PBE-1.2.upf
ATOMIC_POSITIONS {crystal}
Fe 0.000000  0.000000  0.000000
K_POINTS {automatic}
 20 20 20 0 0 0
```

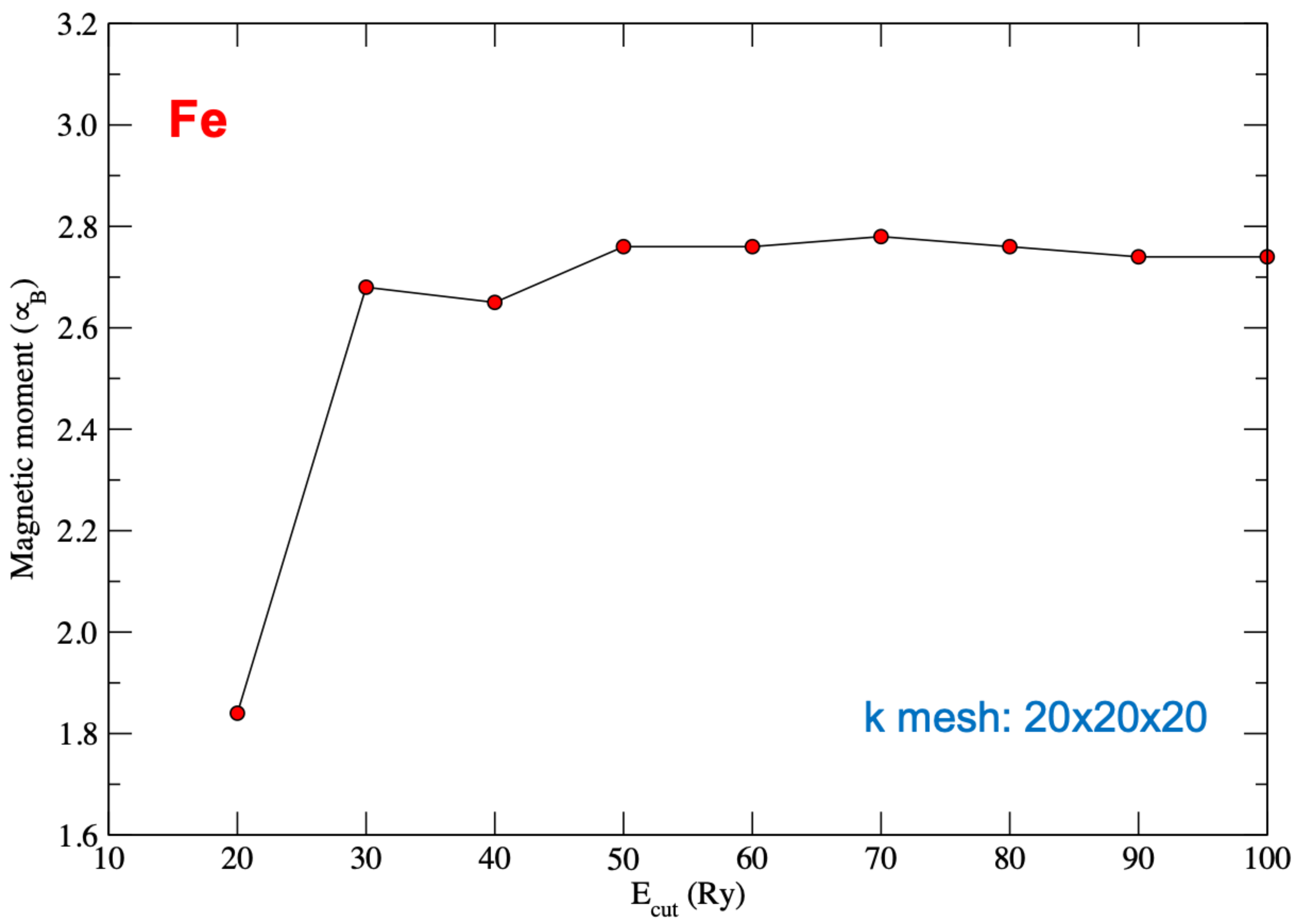
Experimental lattice parameter

It is necessary to perform convergence tests w.r.t ecutwfc

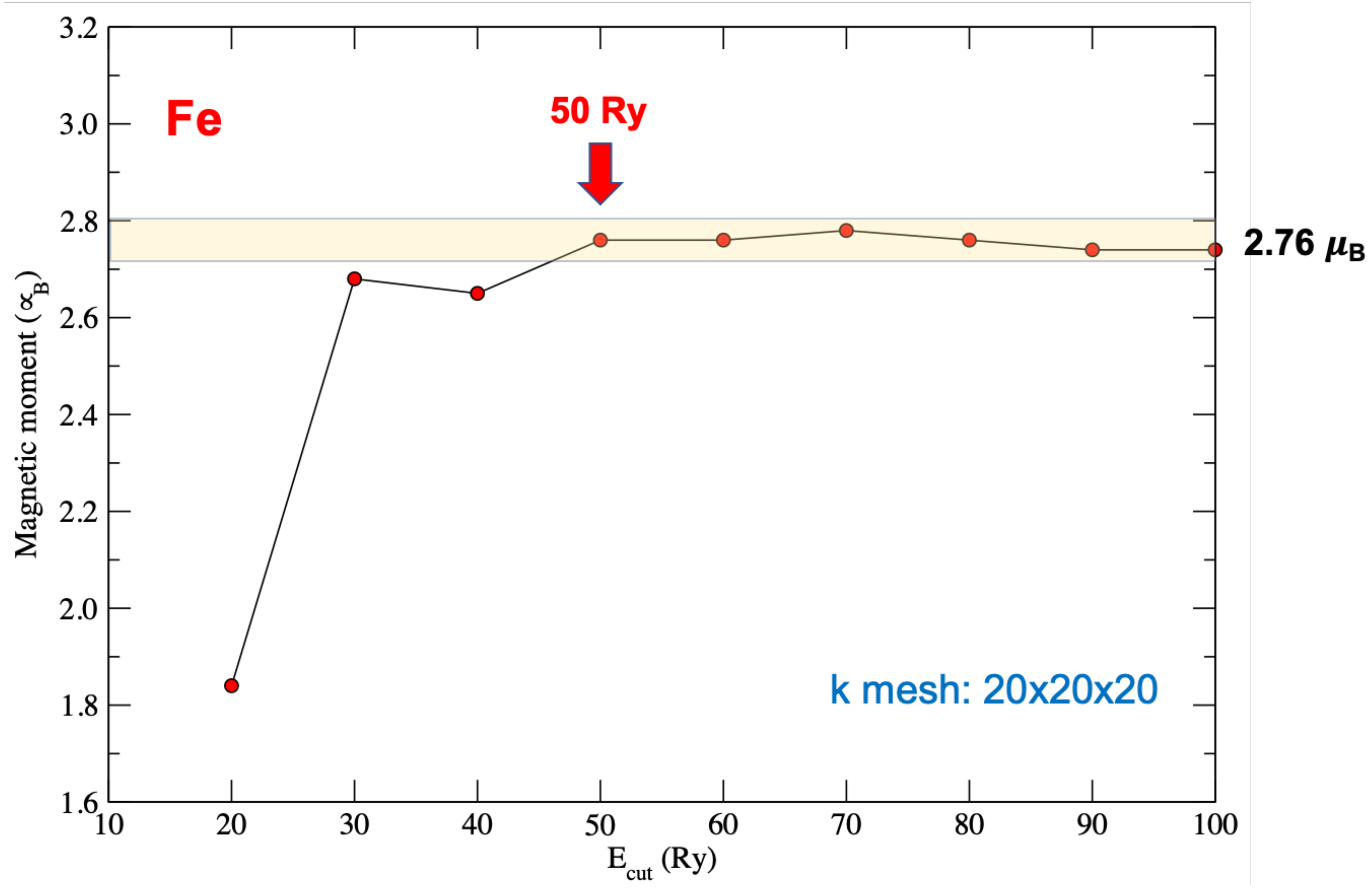
Set up the SCAN functional from the input
Quantum ESPRESSO must be compiled with Libxc

Pseudopotential generated using the PBE functional
SG15 ONCV library (there is no SCAN pseudo for Fe)

SCAN functional & PBE pseudopotential (SG15)



SCAN functional & PBE pseudopotential (SG15)



SCAN functional with different pseudopotentials

Ecut = 50 (Ry), k mesh: 20x20x20

Functional	Pseudopotential	Library	m (μ_B)	
SCAN	PBE	SG15 ONCV	2.76	NLCC=.false.
SCAN	PBE	Pseudo Dojo	2.67	NLCC=.true.
PBE	PBE	SG15 ONCV	2.28	
PBE	PBE	Pseudo Dojo	2.26	

WARNING → (second row)

m_{expt} = 1.98 – 2.13 (μ_B)

SCAN overestimates magnetic moments in itinerant ferromagnets

PBE is in closer agreement with experiments than SCAN for magnetic moments of itinerant ferromagnets

*M. Ekholm et al., Phys. Rev. B **98**, 094413 (2018).*

Different flavors of SCAN

Functional	Pseudopotential	Library	m (μ_B)
SCAN	PBE	SG15 ONCV	2.76
rSCAN	PBE	SG15 ONCV	2.73
r^{2}SCAN	PBE	SG15 ONCV	2.73

m_{expt} = 1.98 – 2.13 (μ_B)

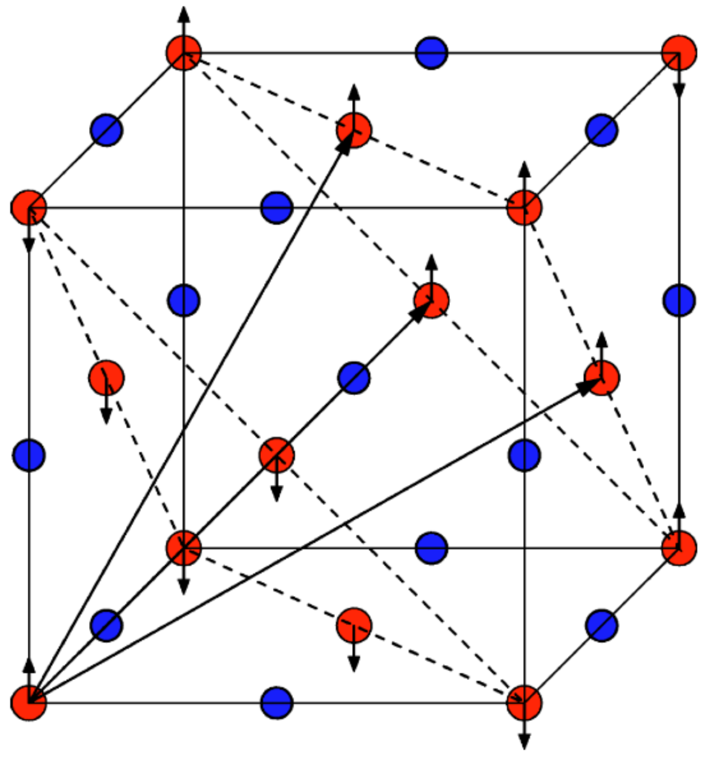
Different flavors of SCAN give very similar magnetic moments for bulk Fe

Exercise 5

Projected density of states of FeO with and without the Hubbard *U* correction

Input file FeO.scf.in

```
&control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 0.0001
U Fe2-3d 0.0001
```



Input file FeO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='FeO'
   pseudo_dir = '../files/pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-9
/
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 0.0001
U Fe2-3d 0.0001
```

Input file FeO.projwfc.in

```
&projwfc
    prefix='FeO'
    outdir='./tmp'
    ngauss = 0,
    degauss = 0.005,
    Emin = -15.0,
    Emax =  30.0,
    DeltaE = 0.01
 /
```

Gaussian broadening for PDOS

Value of the Gaussian broadening (in Ry)

Minimum and maximum energy for the plot (in eV)

Energy grid step

Gnuplot script: plot_pdos.gp

Inspect the script. It aims at plotting Fe-3d states (majority spin and minority spin) and O-2p states.

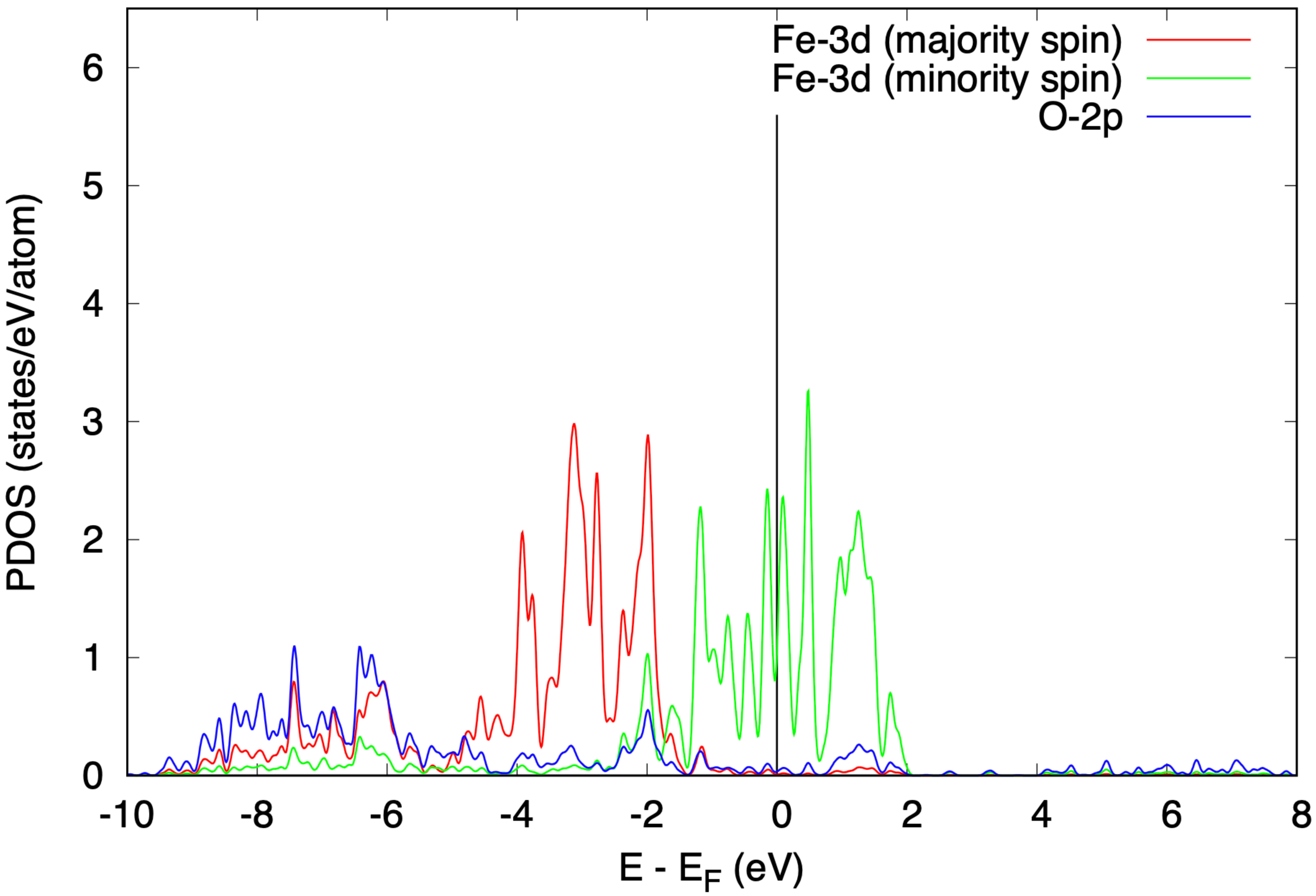
PDOS is shifted such that the Fermi energy corresponds to zero of energy.

After running the script, visualize the file FeO_PDOS.eps.

Run the calculations

```
pw.x < FeO.scf.in |tee FeO.scf.out

pw.x < FeO.nscf.in |tee FeO.nscf.out

projwfc.x < FeO.projwfc.in |tee FeO.projwfc.out

gnuplot plot_pdos.gp

evince FeO_PDOS.eps
```

PDOS using DFT (PBEsol)



DFT predicts FeO to be metallic (this is wrong)

Experimentally FeO is insulating

Let's try DFT+U

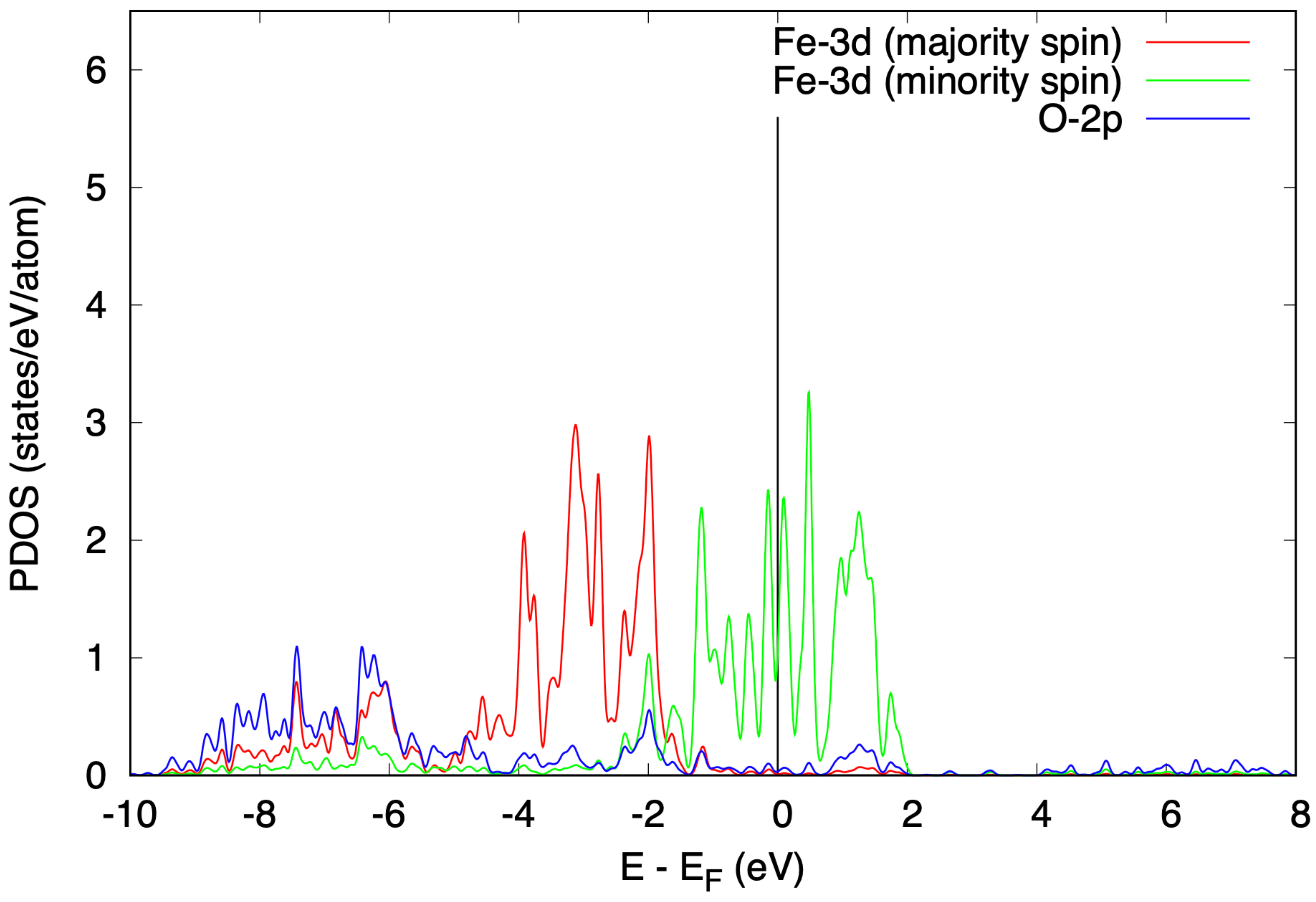
Input file FeO.scf.in

```
&control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

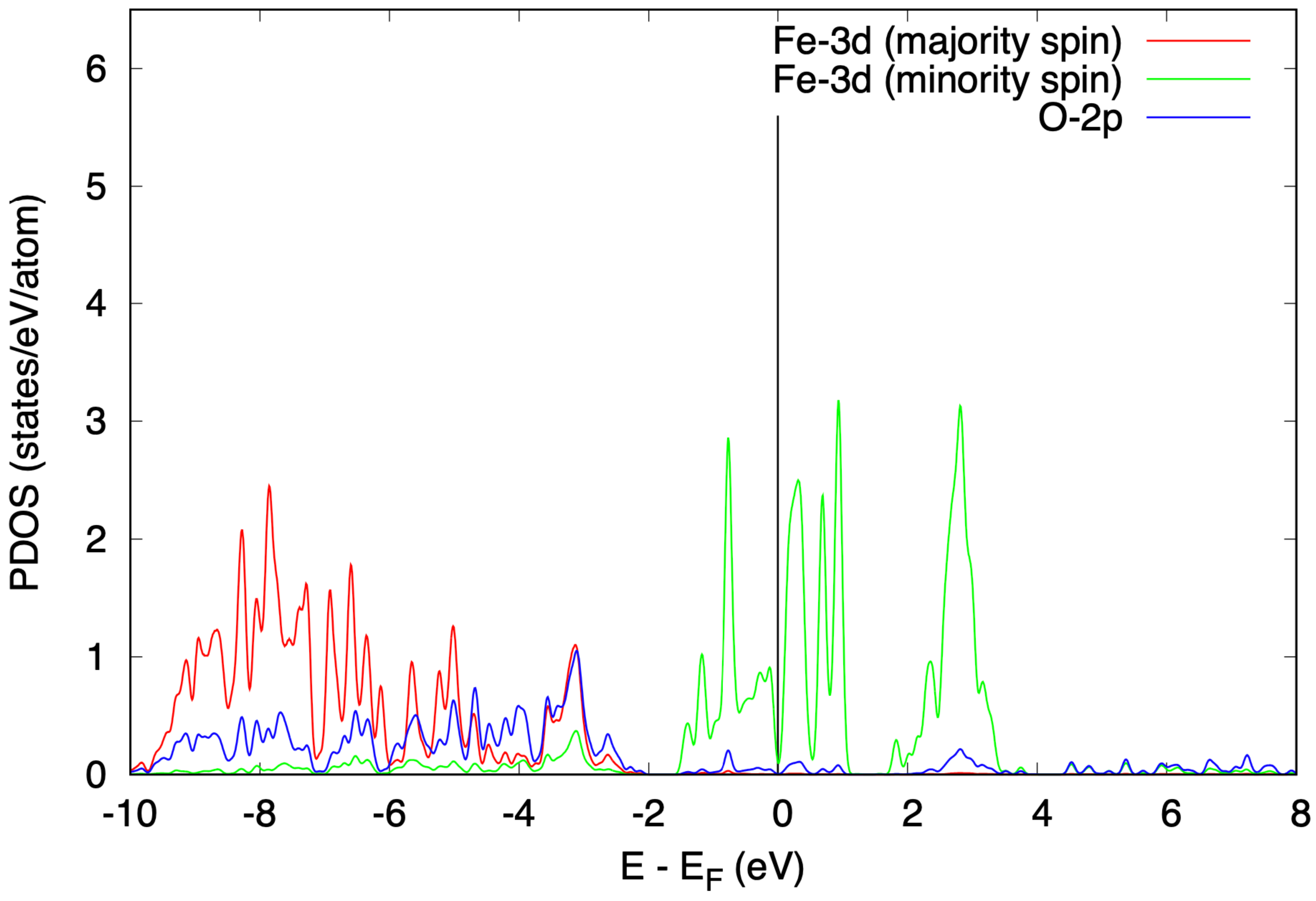
Input file FeO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='FeO'
   pseudo_dir = '../files/pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-9
/
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

PDOS using DFT



PDOS using DFT+*U*



Even a standard DFT+*U* calculation does not manage to open a gap in FeO

Output file FeO.scf.out from the DFT+*U* calculation

After the 1st SCF iteration:

```
=================== HUBBARD OCCUPATIONS ===================
------------------------ ATOM    1 ------------------------
Tr[ns(  1)] (up, down, total) =   4.98138  0.94209  5.92348
Atomic magnetic moment for atom   1 =   4.03929
SPIN  1
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
SPIN  2
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
------------------------ ATOM    2 ------------------------
Tr[ns(  2)] (up, down, total) =   0.94168  4.98133  5.92301
Atomic magnetic moment for atom   2 =  -4.03965
SPIN  1
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
SPIN  2
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
```

Output file FeO.scf.out from the DFT+*U* calculation

After the 1st SCF iteration:

```
=================== HUBBARD OCCUPATIONS ===================
------------------------ ATOM    1 ------------------------
Tr[ns(  1)] (up, down, total) =   4.98138  0.94209  5.92348
Atomic magnetic moment for atom   1 =   4.03929
SPIN  1
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
SPIN  2
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
------------------------ ATOM    2 ------------------------
Tr[ns(  2)] (up, down, total) =   0.94168  4.98133  5.92301
Atomic magnetic moment for atom   2 =  -4.03965
SPIN  1
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
SPIN  2
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
```

starting_ns_eigenvalue(5,2,1) = 1.0

starting_ns_eigenvalue(5,1,2) = 1.0

Output file FeO.scf.out from the DFT+U calculation

After the 1st SCF iteration:

```
=================== HUBBARD OCCUPATIONS ===================
------------------------ ATOM    1 ------------------------
Tr[ns(  1)] (up, down, total) =   4.98138  0.94209  5.92348
Atomic magnetic moment for atom   1 =   4.03929
SPIN  1
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
SPIN  2
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
------------------------ ATOM    2 ------------------------
Tr[ns(  2)] (up, down, total) =   0.94168  4.98133  5.92301
Atomic magnetic moment for atom   2 =  -4.03965
SPIN  1
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
SPIN  2
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
```

starting_ns_eigenvalue(5,2,1) = 1.0

5th eigenvalue

spin index

atomic type

starting_ns_eigenvalue(5,1,2) = 1.0

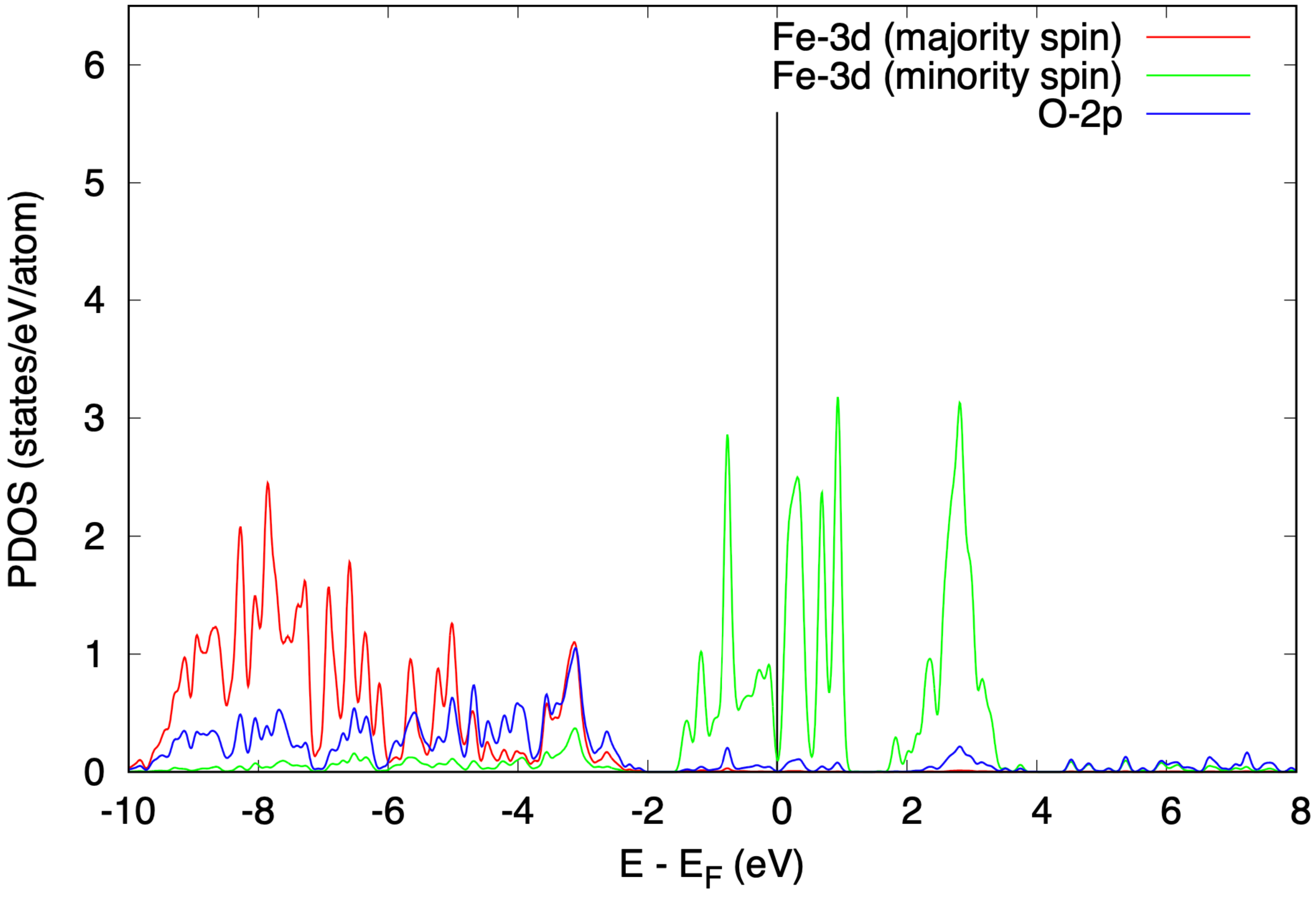
Input file FeO.scf.in

```
 &control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
    starting_ns_eigenvalue(5,2,1) = 1.0
    starting_ns_eigenvalue(5,1,2) = 1.0
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

Input file FeO.nscf.in

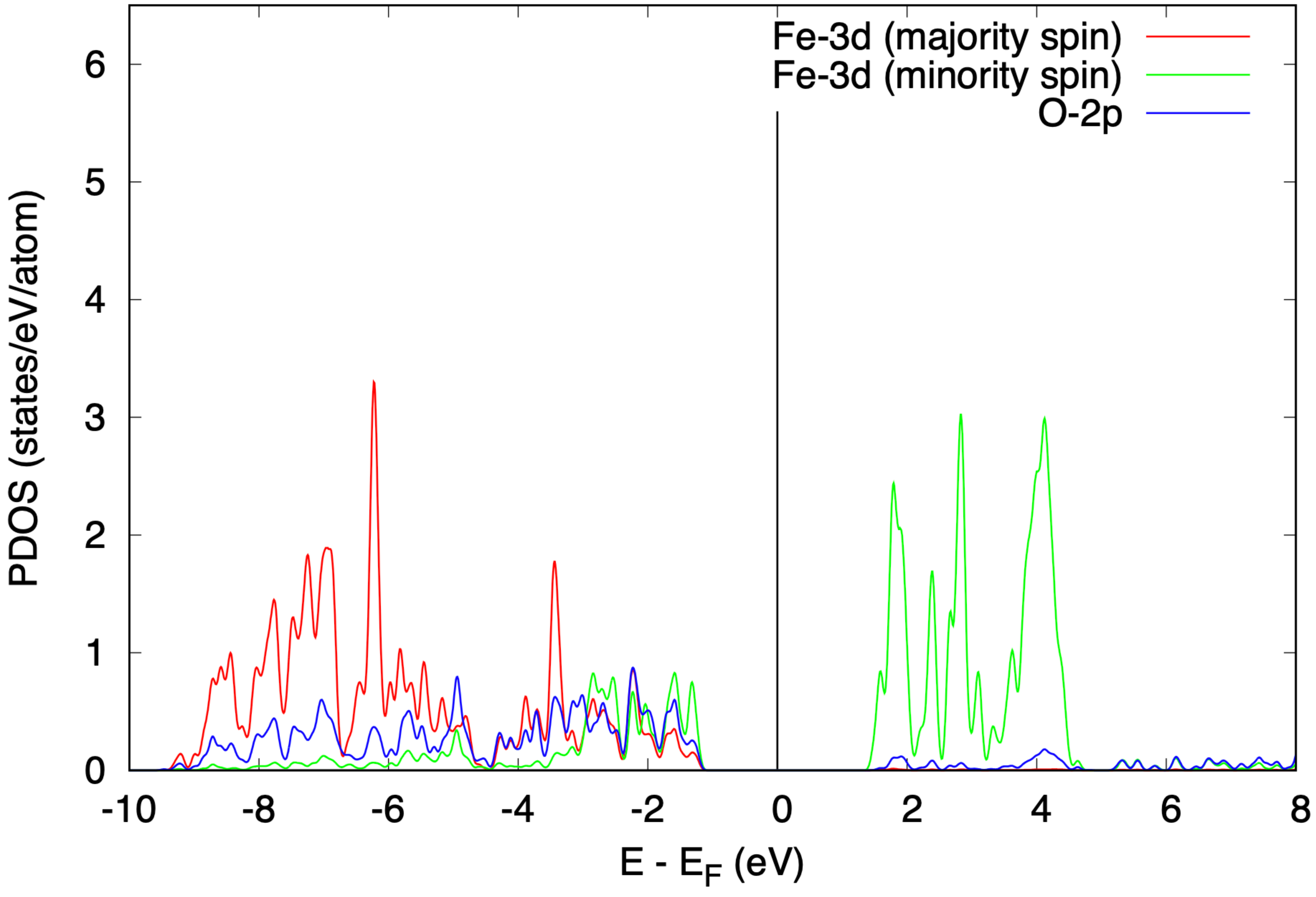
```
 &control
    calculation='nscf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
    starting_ns_eigenvalue(5,2,1) = 1.0
    starting_ns_eigenvalue(5,1,2) = 1.0
    nbnd = 40
 /
 &electrons
    conv_thr =  1.d-9
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

PDOS using DFT+*U*



E_{TOT} = -735.372 Ry

PDOS using DFT+*U* (with starting_ns_eigenvalue)



E_{TOT} = -735.480 Ry

DFT+*U* with starting_ns_eigenvalue opens the band gap in FeO

DFT+*U* band gap: 2.76 eV Experimental gap: 2.4 eV